From Pregnancy Care to Chronic Care: Multi-Drug Evidence on Medicaid's 12-Month Postpartum Extension

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Abstract

Background. Section 9812 of the American Rescue Plan Act of 2021 gave states the option to extend Medicaid postpartum coverage from 60 days to 12 months. By late 2024, 46 states and the District of Columbia had adopted this extension, yet nationally representative causal evidence on outpatient contraceptive utilization remains limited, and the concurrent Medicaid unwinding complicates policy evaluation during the same period.

Methods. We constructed a balanced state-quarter panel of 51 jurisdictions over 36 quarters (N = 1,836) combining Medicaid State Drug Utilization Data (SDUD), which capture pharmacy-dispensed prescriptions, with public-use Medicaid Provider Spending by HCPCS data released by HHS/CMS, which aggregate outpatient and professional Medicaid claims by billing- and servicing-provider NPI, HCPCS code, and month and capture HCPCS- and CPT-billed LARC procedures. We merged both sources with CMS Medicaid Budget and Expenditure System enrollment data. Exploiting staggered state adoption of the 12-month extension, we estimated two-way fixed effects (TWFE), Gardner (2022) two-stage difference-in-differences (did2s), and local projections difference-in-differences (LPDID) models across pharmacy, outpatient procedure, and combined channels. We then used extended drug-category analyses to diagnose the unwinding confound and re-estimated triple-difference (DDD) specifications on transformed relative scales using actual negative control drugs (statins, BPH medications, gout medications, and Alzheimer's drugs).

Results. Standard estimates showed no detectable outpatient LARC effect in any channel: the combined estimate was -0.055 per 1,000 enrollees ($p = 0.604$), the pharmacy-only estimate was -0.010 ($p = 0.915$), and the outpatient-procedure estimate was -0.119 ($p = 0.140$). Negative control drugs with no postpartum clinical connection nonetheless showed negative TWFE coefficients, indicating that the 2023-2024 Medicaid unwinding acts as a broad confound. After re-estimating the LARC-specific DDD on a relative scale using actual negative controls, the primary pooled estimate was modestly positive but imprecise (TWFE log1p DDD = +0.094, $p = 0.095$; did2s = +0.109, $p = 0.078$). Alternative normalizations attenuated the estimate further. Power analysis confirmed that the raw TWFE design could not detect effects of the magnitude found by Rodriguez et al. (2024) after denominator dilution (MDE = 0.255 vs. benchmark = 0.185 per 1,000 total enrollees).

Results . Under corrected treatment dates (NJ and VA SPA 2022, FL 1115 2024; only AR and WI never-treated through 2024Q4), the headline LARC null persists: combined LARC TWFE = -0.062 per 1,000 enrollees ($p = 0.58$); did2s = -0.139 ($p = 0.21$); LARC removal claims (CPT 58301, 11982, 11983) are also null. A long-panel state-quarter FE DDD that fully absorbs the unwinding shock yields LARC log1p coefficient +0.082 ($p = 0.19$); restricted to pre-2023Q1 it collapses to -0.004 ($p = 0.93$). The residual relative-scale positive signal is concentrated entirely in the post-unwinding window. **In contrast, multiple non-LARC outcomes from the same public Medicaid utilization data show statistically detectable positive effects after the policy.** Two HCPCS-based primary-care outcomes are positive on raw TWFE : postpartum blood-pressure monitoring (HCPCS A4670) +0.140 per 1,000 enrollees ($p = 0.001$, +74% off baseline 0.19); established-patient primary-care visits (CPT 99214) +21.6 per 1,000 ($p = 0.024$, +11%). On the SDUD pharmacy side, an expanded 29-category mechanism analysis using state-quarter FE DDD against pooled negative controls reveals a coherent positive signal in chronic disease, harm reduction, severe mental illness, and infectious-disease continuity: insulin (DDD +0.343, $p = 0.006$), naloxone (+0.156, $p = 0.016$), antipsychotics (+0.067, $p = 0.022$), mood stabilizers (+0.065, $p = 0.021$), UTI antibiotics (+0.103, $p = 0.031$), HIV ART (+0.113, $p = 0.034$), and HCV direct-acting antivirals (+0.106, $p = 0.042$). All seven categories with nominal DDD $p < 0.05$ are directionally positive (one-sided exact binomial $p = 0.008$ against the null of equal sign probability); no category is significantly negative. Under Benjamini-Hochberg adjustment across the full 29-category set, no individual category survives FDR $q < 0.05$ (lowest q   0.16), but the consistent positive direction across multiple chronic-care outcomes and the absence of negative DDD-significant categories are meaningful pattern evidence. Pre-treatment event-study Wald tests support a causal reading for insulin, naloxone, antipsychotics, mood stabilizers, UTI antibiotics, and CPT 99214 (Wald $p > 0.10$ in each case); HIV ART is marginal (Wald $p = 0.053$); HCV direct-acting antivirals (Wald $p = 0.023$) and HCPCS A4670 BP monitoring (Wald $p < 0.001$) fail pre-trend testing and are interpreted as consistent with but not separately identified from secular trends. Outpatient LARC, broad short-acting contraception, broad antidepressants, postpartum-compatible antihypertensives, levothyroxine, and SSRI/SNRI categories are all DDD-null.

Conclusions. In public aggregate Medicaid utilization data, 12-month postpartum coverage extensions are not associated with detectable increases in outpatient LARC utilization or LARC removals. They are, however, associated with substantial increases in postpartum hypertension monitoring and primary-care continuity, and directionally positive but imprecise effects on behavioral health utilization. The Medicaid unwinding materially confounds raw post-2022 treated-control comparisons; later-adopting states experienced larger enrollment declines (Pearson $r = -0.39$ between adoption quarter and peak-to-2024Q3 enrollment change, $p = 0.005$). Public aggregate Medicaid utilization data can detect broad shifts in chronic disease management and primary-care continuity after coverage expansions, but they are limited for estimating postpartum-specific contraceptive effects without postpartum-specific denominators or inpatient delivery claims.

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1. Introduction

This paper makes three claims. First, the 12-month postpartum Medicaid coverage extension authorized by Section 9812 of the American Rescue Plan Act of 2021 plausibly improves postpartum access to a broad range of Medicaid-financed services — contraceptive care, postpartum visits, behavioral health, chronic disease management — by removing the 60-day eligibility cliff. Second, outpatient LARC utilization is one measurable channel for that broader access, but it is not a welfare metric and not necessarily the primary policy goal; a patient-centered framework requires attention to the full range of methods, the ability to decline contraception, and access to device removal or method switching. Third, evaluating any 2022-2024 Medicaid policy change requires confronting the Medicaid unwinding — the post-March-2023 resumption of eligibility redeterminations — which acts as a broad common shock that contaminates raw treated-versus-comparison contrasts.

The remainder of Section 1 develops the institutional and clinical context for these claims.

1.1 The Postpartum Coverage Crisis

Maternal mortality in the United States is among the highest in the developed world, and unlike most peer nations, the rate has been rising. Between 2018 and 2021, the maternal mortality rate increased from 17.4 to 32.9 deaths per 100,000 live births, with stark racial disparities: non-Hispanic Black women experienced mortality rates approximately 2.6 times those of non-Hispanic White women (NCHS, 2023). One-third of pregnancy-related deaths occur between one week and one year postpartum, yet federal law historically required states to provide pregnancy-related Medicaid coverage through only 60 days after delivery (MACPAC, 2023). This temporal mismatch – between the period of greatest medical vulnerability and the period of guaranteed coverage – has been the subject of sustained policy attention.

Medicaid is the single largest payer for maternity care in the United States, financing approximately 42% of all births (Martin et al., 2023). For low-income women, Medicaid often provides the only pathway to comprehensive prenatal, delivery, and postpartum services, including contraceptive care. The program's pregnancy-related eligibility pathway typically extends coverage to women with incomes up to 138-200% of the federal poverty level, but this eligibility has historically terminated 60 days after delivery, creating the "coverage cliff."

The consequences of this coverage cliff are well documented. Daw, Hatfield, Swartz, and Sommers (2017) provided the first nationally representative evidence of postpartum coverage disruption, showing that 55% of women with Medicaid or CHIP coverage at delivery experienced a coverage gap within six months of childbirth. This churn is not merely an administrative inconvenience; it disrupts continuity of care during a period of elevated risks for postpartum depression, cardiovascular complications, substance use relapse, and unintended pregnancy (Gordon et al., 2020). Gordon, Sommers, Wilson, and Trivedi (2020) used a difference-in-differences design comparing Colorado to Utah and found that ACA Medicaid expansion yielded approximately 0.9 additional months of postpartum coverage and 0.52 additional outpatient visits, with larger gains among women with serious delivery complications. Their findings established that broader Medicaid eligibility could mitigate the coverage cliff, though the gains were modest and incomplete.

1.2 Postpartum Contraceptive Access and Reproductive Autonomy

Contraceptive care is a key component of postpartum health care when it supports a person's ability to choose if, when, and how to use contraception. For methods such as IUDs and implants, coverage can matter not only for insertion but also for counseling, follow-up, switching, and removal. CDC's 2024 *U.S. Medical Eligibility Criteria for Contraceptive Use* (USMEC) and accompanying *Selected Practice Recommendations* emphasize noncoercive, patient-centered contraceptive counseling and full-method access; contraceptive uptake of any specific method is not, on its own, a measure of better care.

Long-acting reversible contraceptives (LARC), including intrauterine devices (IUDs) and subdermal implants, are one component of the contraceptive method mix that Medicaid finances. They have low typical-use failure rates and require provider insertion, follow-up, and removal. They are also more clinician-dependent than oral contraceptives, the patch, the ring, or DMPA, which means policy effects on Medicaid-financed LARC utilization reflect a combination of patient demand, provider capacity, billing infrastructure, and counseling practices. An increase in Medicaid-financed LARC claims is best interpreted as increased Medicaid-financed access to one clinically intensive contraceptive option — not as a standalone measure of reproductive well-being.

Disparities in postpartum contraceptive access are pronounced. Thiel de Bocanegra, Braughton, Bradsberry, and colleagues (2017) examined 199,860 California Medicaid births and found that Black women were less likely to receive any postpartum contraception or postpartum care than White women. The coverage cliff may compound inequities in access to patient-centered postpartum care, including contraceptive counseling, method choice, follow-up, and removal services. The historical literature also documents that highly clinician-dependent methods such as LARC have been the focus of coercive contraceptive practices in U.S. Medicaid populations, particularly among Black, Latina, Indigenous, and low-income women, and contemporary clinical guidance therefore frames LARC access in terms of voluntary uptake, full-method choice, and unconstrained access to removal — not utilization rates as such.

1.3 The ARPA Section 9812 Option

Section 9812 of the American Rescue Plan Act (ARPA) of 2021 created a state plan amendment (SPA) option allowing states to extend Medicaid postpartum coverage from 60 days to 12 months, effective April 1, 2022. CMS released implementation guidance in December 2021 (SHO #21-007), and the Consolidated Appropriations Act of 2023 made this option permanent. By late 2024, 46 states and the District of Columbia had adopted the extension, with Illinois implementing as early as July 2021 via an 1115 waiver and subsequent adoption concentrated in 2022-2023 (KFF, 2024). MACPAC recommended that Congress mandate the extension for all states with 100% federal matching.

This staggered adoption across states and time creates a natural experiment suitable for difference-in-differences identification. The variation in adoption timing reflects a combination of political will, administrative capacity, and pre-existing Medicaid program structure rather than differential trends in contraceptive utilization. However, the adoption window coincides with two major policy shocks – the end of the COVID-19 continuous enrollment provision and the *Dobbs v. Jackson Women's Health Organization* decision – both of which complicate causal inference.

1.4 The Research Question and the Limits of Aggregate Data

Despite rapid policy uptake and its potential implications for maternal and postpartum health, no study has estimated the causal effect of 12-month postpartum Medicaid coverage extensions on outpatient medication utilization using nationally representative aggregate data. The question this paper poses is therefore narrower than "did the extension improve postpartum care": it is "what can public aggregate Medicaid utilization data tell us about the extension's effect on Medicaid-financed outpatient contraceptive utilization, and how does the Medicaid unwinding complicate that inference?" Existing evidence addresses related but distinct policies. Steenland, Pace, Sinaiko, and Cohen (2019, 2021) evaluated separate Medicaid reimbursement for immediate postpartum LARC (IPP-LARC) in South Carolina and found increased uptake and decreased short-interval births among adolescents. Rodriguez, Meath, Watson, and colleagues (2024) published the largest multi-state IPP-LARC study in *JAMA Health Forum*, using T-MSIS data on 1,378,885 deliveries across 15 states and finding a 0.74 percentage-point increase in IPP-LARC use. Eliason, Spishak-Thomas, and Steenland (2022) found a 7.0 percentage-point increase in postpartum LARC use from ACA Medicaid expansion. Swartz et al. (2025) used individual-level North Carolina Medicaid data to show increased postpartum contraceptive utilization after the extension, but this single-state pre-post comparison cannot isolate the causal effect.

The 12-month postpartum extension is a different policy lever from any of these: it extends the *duration* of eligibility rather than reforming reimbursement rules (as with IPP-LARC policies) or expanding *initial* eligibility (as with ACA expansion). Several null findings temper expectations. Pace, Steenland, Engel, and Sinaiko (2021) found no significant per-capita LARC effect from Medicaid expansion despite absolute increases in prescriptions. Law et al. (2016) found no immediate LARC uptake increase after the ACA contraceptive mandate. These findings suggest that coverage alone may be insufficient when supply-side barriers – including provider training, device stocking, and scheduling logistics – persist.

1.5 Contributions

This paper makes four contributions to the literature on Medicaid coverage policy and postpartum contraceptive access.

First, we develop a multi-data-source framework combining Medicaid State Drug Utilization Data (SDUD) and the public-use Medicaid Provider Spending by HCPCS dataset released by HHS/CMS to assess postpartum medication utilization across pharmacy and outpatient procedure channels. The claims upgrade materially improves measurement of outpatient LARC procedures and yields a cleaner test of whether the raw SDUD null is merely a missing-channel artifact. To our knowledge, this is the first study to combine these data sources for a contraceptive policy evaluation, and the multi-channel framework is portable to other Medicaid drug utilization studies.

Second, we identify the 2023-2024 Medicaid unwinding as a critical confound in contemporaneous Medicaid policy evaluations. Negative control drugs with no postpartum clinical relevance show negative TWFE coefficients, indicating that raw post-2022 contrasts are contaminated by a broad enrollment shock. This finding has implications well beyond the current study: any difference-in-differences evaluation of Medicaid policies adopted during 2022-2024 must contend with the unwinding as a simultaneous shock.

Third, once the LARC-specific triple-difference (DDD) is re-estimated on a defensible relative scale using the paper's actual negative controls, the relative-scale signal is directionally positive but imprecise (TWFE log1p DDD: +0.094, p = 0.095; did2s: +0.109, p = 0.078), and alternative normalizations attenuate further. The strongest causal claim is therefore not that the extension clearly increases outpatient LARC utilization, but that raw nulls are partly confounded by the unwinding while the residual positive evidence is modest and sensitive to specification — what aggregate public data can show without postpartum-specific denominators is bounded.

Fourth, formal power analysis confirms that the raw TWFE design cannot detect effects of the magnitude found by Rodriguez et al. (2024) after denominator dilution: the minimum detectable effect for the pharmacy channel is 0.255 per 1,000 total enrollees (26% of baseline), which exceeds the Rodriguez benchmark of 0.185. Together, these results imply that stronger first-stage evidence likely requires inpatient IPP-LARC claims or postpartum-specific denominators.

Our findings carry implications for researchers and policymakers. The identification of the unwinding as a pervasive confound — evidenced by significant negative coefficients for negative-control drugs with no postpartum relevance — provides a cautionary finding for all contemporaneous Medicaid policy evaluations and a diagnostic template for common shocks during the redetermination period. The re-estimated LARC DDDs are directionally positive but modest and imprecise; they are best read as suggestive relative evidence, not proof of a large outpatient utilization response. Back-of-envelope denominator adjustment illustrates the scale of attenuation: postpartum-eligible women represent approximately 1.9% of total Medicaid enrollment, yielding a scaling factor of approximately 57x — meaning aggregate per-enrollee designs are structurally underpowered for postpartum-specific effects. We do not interpret the null as evidence that the extension is ineffective; rather, that aggregate public data are not the right instrument for measuring it.

The remainder of this chapter proceeds as follows. Section 2 provides institutional background and reviews the relevant literature. Section 3 describes the data sources and sample construction. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 discusses findings, and Section 7 concludes.

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2. Background and Literature Review

2.1 The Postpartum Coverage Cliff and Its Consequences

The foundational policy problem motivating this study is the historically limited duration of pregnancy-related Medicaid coverage. Under federal law prior to ARPA, states were required to provide Medicaid coverage for pregnant women during pregnancy and through 60 days postpartum, after which coverage terminated unless the woman qualified through another eligibility pathway (e.g., adult expansion, disability). The 60-day cutoff was established by the Omnibus Budget Reconciliation Act of 1986 and had not been updated for over three decades despite substantial changes in the understanding of postpartum health needs.

The clinical rationale for longer postpartum coverage is compelling. ACOG has emphasized that the postpartum period is increasingly understood as extending to 12 months, with ongoing needs for chronic disease management, mental health treatment, substance use disorder care, and contraceptive services (ACOG, 2018). The CDC's Pregnancy Mortality Surveillance System documented that approximately one-third of pregnancy-related deaths occur between 43 days and one year postpartum, falling entirely outside the guaranteed coverage window under the 60-day rule.

Daw et al. (2017) provided the first nationally representative quantification of the coverage cliff's consequences. Using longitudinal MEPS data, they showed that coverage churn around pregnancy was pervasive: 55% of women with public insurance at delivery experienced a gap within six months. The coverage disruption was not randomly distributed; it disproportionately affected women of color, women with lower educational attainment, and women in states that had not expanded Medicaid under the ACA. This differential exposure to coverage loss has important implications for health equity, as the populations most likely to lose coverage are also those with the highest rates of adverse postpartum outcomes. Johnston, McMorrow, Thomas, and Kenney (2020) similarly documented that Medicaid expansion reduced uninsurance among new mothers in poverty, but significant coverage gaps persisted.

The COVID-19 pandemic created a natural experiment that powerfully demonstrated the magnitude of coverage losses under the 60-day rule. The Families First Coronavirus Response Act (FFCRA) continuous enrollment provision increased the rate of continuous 12-month postpartum Medicaid enrollment from 59.3% to 90.7%, corresponding to approximately 430,000 fewer people losing postpartum coverage (Gordon et al., 2024). This finding revealed that roughly 40% of Medicaid-covered births previously resulted in coverage loss before 12 months postpartum – a staggering rate of disruption that the 12-month extension was designed to address.

2.2 Section 9812 and State Adoption

Section 9812 represented the most significant legislative change to postpartum Medicaid coverage since the original 60-day mandate. CMS implementation guidance (SHO #21-007) specified that the extension applied to all pregnancy-related eligibility categories and provided the **full Medicaid benefit package** — i.e., the full set of state-plan or 1115-waiver services available to categorically needy adults, not merely pregnancy-related services. CMS guidance is explicit that coverage limited to pregnancy-related services does not satisfy the 12-month-extension definition. State adoption was remarkably rapid by the standards of optional Medicaid provisions: the modal adoption quarters were 2022 Q3 and 2022 Q4, each with 9 states. By the end of the study window, the only jurisdictions that had not yet implemented full 12-month coverage were Arkansas and Wisconsin; New Jersey and Virginia adopted via SPA in 2022 and Florida adopted via Section 1115 waiver in 2024 (KFF, 2024; CMS SPA tracker). The current panel codes NJ, VA, and FL as never-treated through 2024Q4; this coding is being audited and revised (see the treatment-coding appendix), and all reported estimates should be re-run on corrected treatment dates before submission.

Adoption of the 12-month eligibility pathway does not ensure equal implementation. States varied in outreach, MCO contract requirements, eligibility-system readiness, provider billing guidance, and contemporaneous unwinding procedures. We therefore interpret adoption as an intent-to-expand-coverage policy exposure, not as uniform implementation of postpartum care delivery. Some states, most notably Illinois, implemented earlier through Section 1115 waivers. The compressed adoption window is important for identification: it provides substantial cross-state variation in treatment timing but limits the power of heterogeneous treatment effect analyses by adoption cohort. ASPE issue briefs estimated that hundreds of thousands of additional person-months of coverage are generated annually by the extension (Gordon et al., 2023).

We make no welfare claim about LARC uptake. The conceptual framework in this paper runs **policy adoption → coverage continuity/certainty → Medicaid-financed postpartum access → utilization of outpatient services → health and autonomy-related outcomes**, with multiple potential barriers (provider capacity, billing, MCO rules, patient preferences, inpatient LARC availability, *Dobbs*, supply shocks, and the unwinding) intervening between adoption and observed utilization. LARC uptake is not inherently "good" or "bad"; the welfare interpretation depends on whether patients wanted that method, had alternatives, and had access to follow-up and removal.

The mechanism through which the extension could affect contraceptive utilization operates through the demand side: by extending the period of guaranteed insurance coverage, the extension reduces the effective price of contraceptive services for postpartum women who would otherwise lose coverage at 60 days. For LARC specifically, the extension may matter most for women who do not receive LARC during the delivery hospitalization (immediate postpartum, or IPP-LARC) and instead seek outpatient insertion at a subsequent visit – typically 4 to 8 weeks postpartum, when many LARC insertions occur clinically. Under the 60-day rule, a woman attending a 6-week postpartum visit would have coverage for LARC insertion, but if that visit were delayed even slightly, she could arrive after coverage termination. The 12-month extension provides a substantially longer window for outpatient LARC access, and it also covers the ongoing monitoring visits and potential removal/replacement that LARC requires.

2.3 IPP-LARC Reimbursement Policies

A distinct but related policy lever involves Medicaid reimbursement reforms that unbundle IPP-LARC from the global obstetric fee. This is conceptually different from the coverage extension: IPP-LARC reimbursement addresses a *supply-side billing barrier* at the point of delivery, while the coverage extension addresses a *demand-side eligibility duration* barrier in the outpatient setting. As of October 2023, 45 states and DC had published Medicaid IPP-LARC reimbursement guidance (ACOG, 2023).

Steenland et al. (2021) found that South Carolina's policy increased monthly IPP-LARC adoption, with effects 2.2 times larger for adolescents. Critically, only 43% of facilities offered IPP-LARC post-policy, highlighting implementation barriers beyond reimbursement. Rodriguez et al. (2024) found a 0.74 percentage-point increase in IPP-LARC use (from a 0.54% baseline – over 100% relative increase) across 15 states using T-MSIS data on 1,378,885 deliveries. This provides the most relevant benchmark for policy-induced LARC effects, though the IPP-LARC reimbursement mechanism is distinct from the coverage extension studied here.

Marthey and colleagues (2024) in *Health Services Research* examined downstream effects of state Medicaid IPP-LARC reforms on maternal mental health using BRFSS data (2014-2019) and found that adoption was associated with 5.7%-11.5% reductions in the probability of reporting poor mental health days among low-income mothers. This finding demonstrates that LARC policy effects can extend beyond contraceptive outcomes to broader maternal well-being.

2.4 Insurance Coverage and Contraceptive Use

The broader literature provides the theoretical and empirical foundation for hypothesizing that coverage extensions will increase LARC uptake. For LARC, with high upfront device costs ($800-$1,300 for an IUD) but low marginal costs after insertion, the demand response to insurance is expected to be large at the extensive margin. Carlin, Fertig, and Dowd (2016) found that the ACA's elimination of contraceptive cost sharing increased $0 cost-sharing claims for IUDs from 36.6% to 87.6%. Eliason et al. (2022) found that ACA Medicaid expansion was associated with a 7.0 percentage-point increase in postpartum LARC use. However, the ACA expansion operated through expanding initial eligibility rather than extending duration, and the 7.0 percentage-point effect may represent an upper bound for the coverage extension, which targets women who already had coverage at delivery.

Several studies report null effects, providing important context. Pace et al. (2021) found no significant per-capita LARC effect from Medicaid expansion despite absolute increases in prescriptions, suggesting that per-capita effects can be difficult to detect in aggregate data. Law et al. (2016) found no immediate jump in LARC uptake after the ACA contraceptive mandate. Malone, Essex, Frech, and colleagues (2024) demonstrated that continuity of health insurance affects contraceptive method choice, but the relationship between coverage and LARC is mediated by provider capacity, billing complexity, and patient preferences. These findings suggest that coverage alone may be insufficient when supply-side barriers constrain uptake.

2.5 Racial and Ethnic Disparities in Postpartum Contraceptive Access

The equity dimension of postpartum contraceptive policy is central to the policy motivation. Thiel de Bocanegra et al. (2017) documented that among nearly 200,000 California Medicaid births, Black women were less likely to receive postpartum care, any contraception, and highly effective contraception compared to White women. These disparities operate through multiple channels: differential access to prenatal counseling about postpartum contraception, differential rates of postpartum visit attendance, provider implicit bias in contraceptive counseling, and structural barriers including transportation and childcare.

Eliason (2020) used a DiD design to show that Medicaid expansion was associated with 7.01 fewer maternal deaths per 100,000 live births, with significant reductions among non-Hispanic Black women. While this study focused on mortality rather than contraceptive access, it underscores how postpartum coverage policy intersects with structural racism in maternal health. The 12-month extension has the potential either to narrow or to widen racial disparities in postpartum contraceptive access, depending on whether the barriers facing women of color are primarily financial (coverage-responsive) or non-financial (coverage-independent). This heterogeneity motivates the disaggregated analyses in the planned future research agenda.

2.6 Contemporaneous Policy Shocks

The *Dobbs v. Jackson Women's Health Organization* decision (June 2022) introduced a contemporaneous shock to contraceptive demand. A Becker Friedman Institute working paper (2025) found a 15% increase in LARC insertions in abortion-hostile states immediately after *Dobbs*, though differences were short-lived for most groups, converging by September 2022. If abortion-hostile states were slower to adopt the postpartum extension, the *Dobbs*-induced LARC increase in control states could bias estimates downward. We address this through heterogeneous treatment effect analyses by state abortion policy environment (Section 5).

The most consequential contemporaneous shock for this study is the Medicaid unwinding. The FFCRA continuous enrollment provision expired on March 31, 2023, after which states were permitted to resume eligibility redeterminations and disenroll individuals who no longer met eligibility criteria. CMS estimated that up to 15 million people could lose Medicaid coverage during the redetermination period.

The unwinding is particularly problematic for DiD designs studying 2022-2023 policies because it represents a common shock that differentially affected states based on their pre-unwinding enrollment levels, the pace of their redetermination processes, and the extent to which pandemic-era enrollment gains included individuals who would not have qualified absent the continuous enrollment provision. States that expanded Medicaid under the ACA and adopted the postpartum extension – which substantially overlap – tended to have larger pandemic-era enrollment gains and thus larger unwinding effects. This creates a mechanical correlation between treatment status and the unwinding shock that contaminates raw difference-in-differences comparisons.

The unwinding affects our estimates through at least two channels. First, the shrinking denominator (total Medicaid enrollment) mechanically changes per-capita rates even if the numerator (LARC prescriptions) is unchanged. Second, the composition of the remaining Medicaid population changes as healthier, less medically complex enrollees disenroll first, potentially altering the average utilization intensity. Our extended drug analysis provides direct evidence of this confound by showing that negative control drugs with no postpartum relevance also show negative TWFE coefficients (Section 5.7).

2.7 Econometric Methods for Staggered Adoption

This paper employs modern DiD methods designed for settings with staggered treatment adoption and potentially heterogeneous treatment effects. Goodman-Bacon (2021) showed that the TWFE estimator is a variance-weighted average of all possible 2x2 comparisons in a panel, and that these weights can be negative when treatment effects are heterogeneous or dynamic. In staggered settings, some comparisons use already-treated units as controls, which can produce biased estimates if treatment effects evolve over time. Callaway and Sant'Anna (2021) estimate group-time ATTs that avoid contamination from already-treated comparisons. Gardner (2022) proposes a two-stage procedure (did2s) that estimates fixed effects using untreated observations only. Dube, Girardi, Jorda, and Taylor (2023) develop a local projections approach (LPDID) that provides a flexible, semi-parametric alternative. We implement all three to ensure that our null finding is not an artifact of estimator choice, and we report the Goodman-Bacon decomposition to assess the contribution of potentially problematic comparisons.

2.8 SDUD as a Research Tool

The Medicaid State Drug Utilization Data, maintained by CMS, provides quarterly state-level counts and expenditures for all outpatient drugs paid by Medicaid, identified by NDC. Wen, Hockenberry, Borders, and Druss (2017) demonstrated its utility for tracking state-level prescription drug utilization in response to policy changes. SDUD captures only pharmacy-dispensed drugs and does not capture device-based LARC provision during inpatient delivery stays (IPP-LARC), because devices bundled into DRG payments are excluded by statute and unbundled IPP-LARC billing flows through facility claims rather than the pharmacy point of sale. States perform HCPCS-to-NDC crosswalks inconsistently. This measurement limitation – which our combined panel with T-MSIS outpatient claims is designed to address – is a central feature of the paper's analytical framework.

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3. Data and Sample Construction

3.1 Data Sources

3.1.1 Medicaid State Drug Utilization Data (SDUD) The outcome data come from the CMS Medicaid State Drug Utilization Data, which reports quarterly state-level counts and expenditures for all outpatient drugs paid by Medicaid, identified by 11-digit National Drug Code (NDC). We downloaded annual SDUD files for 2016-2024 from the CMS Medicaid.gov portal (CMS, 2024). Each record represents a unique state-quarter-NDC-utilization type observation, where utilization type distinguishes fee-for-service (FFSU) and managed care organization (MCOU) claims. We aggregated both to capture the full Medicaid pharmacy benefit.

We identified LARC products using nine NDC codes spanning seven FDA-approved products: four hormonal IUDs (Mirena [2 NDCs], Kyleena, Skyla, Liletta), one copper IUD (Paragard), and two subdermal implants (Nexplanon [2 NDCs, reflecting the 2021 Organon spinoff from Merck], Implanon [discontinued]). We aggregated LARC prescription counts to the state-quarter level.

For the extended drug analysis, we identified NDC codes for eight postpartum-relevant treatment drug categories – short-acting contraceptives, antidepressants (SSRIs/SNRIs), antihypertensives, thyroid medications, diabetes medications, non-benzodiazepine anxiolytics, anticoagulants, and medications for opioid use disorder (MOUD) – and four negative control categories with no clinical connection to postpartum care: statins, BPH medications, gout medications, and Alzheimer's drugs.

3.1.2 Medicaid Enrollment Data Denominators come from the CMS Medicaid Budget and Expenditure System (MBES), which reports monthly state-level Medicaid enrollment counts. We averaged the three monthly values within each calendar quarter to construct quarterly enrollment denominators. The MBES data do not disaggregate enrollment by age or sex; therefore, we use total Medicaid enrollment as the denominator. This choice is consistent with prior SDUD-based research (Wen et al., 2017). It is not "conservative" in any directional sense: a denominator that is far larger than the policy-eligible population dilutes per-enrollee rates and reduces statistical power, but the unwinding shrinks that denominator differentially across states and over time, so a per-enrollee rate constructed from total enrollment can be biased toward zero, away from zero, or in either direction depending on the joint dynamics of numerator and denominator. We address this by reporting numerator-only and Poisson-with-log-enrollment-offset specifications alongside the per-1,000 rate (Section 5).

3.1.3 CMS/HHS Medicaid Provider Spending by HCPCS Public-Use Data To capture LARC procedures delivered outside the pharmacy channel, we use the public-use Medicaid Provider Spending by HCPCS dataset released by HHS/CMS in early 2026. This file aggregates outpatient and professional Medicaid claims (fee-for-service and managed care) and reports, for each unique combination of billing-provider NPI, servicing-provider NPI, HCPCS code, and month/year: the count of distinct beneficiaries, the count of procedures, and the total paid amount. The dataset **excludes institutional claims (no inpatient hospital records) and excludes prescription drug claims**. As described by KFF, it is the most granular Medicaid utilization dataset that has been publicly released in over a decade, but its unit of observation is provider-provider-HCPCS-month, not state-quarter; we constructed our state-quarter aggregates by joining each NPI to its NPPES-listed practice-location state and summing within state and quarter.

State assignment in the resulting state-quarter panel is therefore a **provider-state**, not a beneficiary-state, measure. Specifically, it reflects the practice location of the **servicing-provider** NPI (with billing-provider state as a sensitivity check). This is an important caveat for state-policy designs: a state's outpatient claims volume reflects services delivered by providers practicing in that state, which need not coincide with the state of beneficiary residence (e.g., border-county care, telehealth across state lines).

We identified LARC procedures billed through HCPCS J-codes (J7296, J7297, J7298, J7300, J7301, J7307) and CPT procedure codes for IUD and implant insertion (58300, 11981), removal (58301, 11982), and replacement (11983). Like SDUD, this dataset does not capture inpatient IPP-LARC, which is billed through institutional facility claims (UB-04 type 13x) excluded from the public file. The state-quarter panel constructed from this dataset has 1,357 observations with non-missing claims.

Several data quality limitations warrant discussion. First, cell suppression at the provider-month-HCPCS level (rows with fewer than 12 beneficiaries dropped per CMS suppression policy) systematically excludes small states with low procedure volumes — the missingness is non-random and correlated with state size, treatment status, and managed-care reporting completeness. Second, the 26% missing rate at the state-quarter level necessitates zero-imputation for the combined panel; this can bias estimates either toward or away from the null depending on whether suppressed cells are concentrated in treated or comparison states. We assess sensitivity by estimating on the non-imputed subsample (Table C5). Third, the data lack patient demographic information, so we cannot restrict to postpartum women. Fourth, as a recently released public-use file, the dataset has not been independently validated against the restricted CMS T-MSIS/TAF research files.

We constructed three outcome measures: (1) pharmacy-dispensed LARC per 1,000 enrollees (SDUD), (2) outpatient LARC procedures per 1,000 enrollees (T-MSIS), and (3) combined LARC utilization per 1,000 enrollees (summing pharmacy and procedure counts). The combined measure provides the most comprehensive available estimate of outpatient LARC utilization, though it still excludes inpatient IPP-LARC.

3.1.4 National Vital Statistics System (NVSS) The planned reduced-form analysis of interpregnancy intervals requires restricted-use natality data from the National Vital Statistics System (NVSS), maintained by the National Center for Health Statistics (NCHS). The NVSS restricted natality files contain state-level birth records with detailed maternal characteristics including previous birth dates (allowing computation of interpregnancy intervals), maternal age, race/ethnicity, parity, insurance status at delivery, and gestational age. At the time of writing, the restricted NVSS natality files for 2022-2024 have not yet been obtained, and the reduced-form IPI analysis remains a planned future component. The current paper focuses exclusively on the first-stage analysis of LARC and other medication utilization.

3.1.5 State Policy Data Treatment timing comes from the KFF Medicaid Postpartum Coverage Extension Tracker (KFF, 2024), cross-referenced with CMS SPA approval records and the National Academy for State Health Policy tracker. We assigned treatment to the first full calendar quarter following CMS approval. Treatment dates range from 2021 Q3 (Illinois) through 2024 Q3 (Nevada). Two states — Arkansas and Wisconsin — are never-treated through 2024 Q4. New Jersey and Virginia adopted via SPA in 2022; Florida adopted via Section 1115 waiver in 2024. The full per-state treatment-coding table is in the online appendix. The design is identified primarily by not-yet-treated comparisons across staggered adoption cohorts, supplemented by two never-treated states (AR, WI) for late-period anchoring; we report sensitivity to four treatment-quarter definitions (approval, effective, first-full-quarter, and one-quarter lag) in the appendix.

3.2 Sample Construction

The primary analytic sample is a balanced state-quarter panel comprising 51 jurisdictions (50 states plus the District of Columbia) observed over 36 quarters (2016 Q1 through 2024 Q4), yielding 1,836 observations for the pharmacy channel. Each observation records total LARC prescriptions, prescriptions by device type (IUD vs. implant) and utilization type (FFS vs. MCO), extended drug category counts, total Medicaid enrollment, and treatment status indicators. The outpatient claims panel includes 1,357 state-quarter observations with non-missing procedure counts. The combined panel (N = 1,836) sums pharmacy prescriptions and outpatient procedure counts, with procedure counts set to zero for state-quarters without public HCPCS data.

The descriptive statistics in Table 1 (current draft) report 1,323 treated state-quarter observations and 180 never-treated state-quarter observations, totaling 1,503. This is internally inconsistent with the 1,836-observation balanced panel: the 1,323 figure represents *post-adoption* treated state-quarter observations only, and excludes 333 *pre-adoption* treated state-quarter observations. The corrected mutually-exclusive grouping is: pre-adoption treated state-quarter obs (N = 333), post-adoption treated state-quarter obs (N = 1,323), never-treated state-quarter obs (N = 180), summing to 1,836. Under the corrected treatment coding (Section 3.1.5), the never-treated count drops to 72 (AR + WI × 36 quarters) and the treated state-quarter counts adjust accordingly. Table 1 will be rebuilt with mutually exclusive groups (state count and observation count per row) and a treatment-timing figure with cohort sizes before submission.

We use a balanced panel rather than an unbalanced panel to avoid compositional changes in the state sample over time. For the combined panel, we assess sensitivity to the zero-imputation decision by estimating models on the T-MSIS subsample only (Table C5). We aggregate across all utilization types (FFS and MCO) for the primary analysis, since the postpartum extension covers women in both delivery systems. We report FFS-only and MCO-only estimates as heterogeneity analyses.

Negative control drugs were selected to satisfy two criteria: (a) no clinical connection to postpartum care and (b) sufficient prescription volume for stable rate estimation. Statins (used for hyperlipidemia), BPH medications (used for benign prostatic hyperplasia, a male-only condition), gout medications, and Alzheimer's drugs satisfy both criteria. The choice of these specific controls allows us to test whether observed changes in drug utilization rates reflect postpartum-specific policy mechanisms or broader Medicaid enrollment dynamics.

The primary outcome is LARC utilization per 1,000 Medicaid enrollees, measured separately for each channel. For the pharmacy channel, treated states average 0.954 per 1,000 (SD = 0.551) compared with 0.722 per 1,000 (SD = 0.496) in never-treated states. For outpatient procedures, the mean rate is 0.422 per 1,000 (SD = 0.660) among treated states and 0.377 per 1,000 (SD = 0.292) among never-treated states. The combined rate averages 1.242 per 1,000 (SD = 0.751) in treated states and 1.015 per 1,000 (SD = 0.546) in never-treated states (Table C1). IUDs account for approximately 82% of all LARC prescriptions in SDUD, with implants accounting for the remainder. The outpatient claims channel accounts for approximately 16% of total LARC volume among treated states and 27% among never-treated states.

An important feature of the outcome measure is denominator dilution: postpartum-eligible women represent only approximately 1.9% of total Medicaid enrollment, yielding a scaling factor of roughly 57x. This dilution is the primary source of limited statistical power (Section 5.10). The choice is conservative – it cannot produce false positives – but can produce false negatives if the true effect is moderate in magnitude but small relative to total enrollment.

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4. Empirical Strategy

4.1 Identification Strategy

The identification strategy exploits staggered adoption of the 12-month extension across states between 2021 Q3 and 2024 Q3. The key identifying assumption is that, absent the extension, treated and comparison states would have followed parallel trends in LARC utilization. This assumption is supported by the event study (Section 5.4). Identifying variation comes from (1) treated vs. never-treated states and (2) early vs. later adopters before the latter are treated. The CS and did2s estimators use both sources of variation.

Several threats to identification warrant discussion. First, adoption is not randomly assigned; states with more generous Medicaid programs may have adopted earlier. We address this through the event study and pre-trends analysis. Second, the compressed adoption window limits variation in treatment timing. Third, the concurrent unwinding and *Dobbs* create common shocks that may differentially affect treated and comparison states. The DDD framework and heterogeneity analyses address these concerns, though imperfectly.

4.1A Treatment Definition and Sensitivity

We define treatment using the state-specific effective date of 12-month postpartum coverage, not announcement or approval date. The primary treatment indicator turns on in the first full quarter in which coverage was operational. We evaluate sensitivity to four alternative definitions in parallel: (a) approval date (first quarter CMS SPA approval), (b) effective date (first quarter effective date listed in the SPA or 1115 waiver), (c) first-full-quarter (the default), and (d) one-quarter lag of the first-full-quarter definition (to allow time for provider operational response). Where adoption is via partial or waiver-based coverage rather than full SPA-based extension, we additionally report estimates excluding those states. The full per-state treatment-coding table is in the online appendix.

4.2 Two-Way Fixed Effects (TWFE)

Our baseline specification is:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot D_{st} + \varepsilon_{st}$$

where y_{st} is the LARC rate per 1,000 enrollees in state s and quarter t, α_s are state fixed effects, γ_t are quarter fixed effects, D_{st} indicates implementation of the extension, and ε_{st} is the error term. Standard errors are clustered at the state level (Bertrand, Duflo, and Mullainathan, 2004). The state fixed effects absorb all time-invariant state-level differences; the quarter fixed effects absorb all national-level trends. The coefficient β is identified from within-state changes coinciding with adoption, relative to comparison states.

We implement the Goodman-Bacon (2021) decomposition. Already-treated-as-control comparisons carry 36% of total weight with a mean estimate of -0.041, while clean comparisons are centered near zero, suggesting slight downward bias in the TWFE estimate and motivating the heterogeneity-robust estimators below.

4.3 Two-Stage Difference-in-Differences (did2s)

The Gardner (2022) did2s estimator estimates fixed effects using only untreated observations in the first stage, then regresses the residualized outcome on the treatment indicator in the second stage. This avoids bias from already-treated units serving as controls. Standard errors use the bootstrap procedure recommended by Gardner (2022).

4.4 Local Projections DiD (LPDID)

Following Dube et al. (2023), LPDID estimates the treatment effect at each post-treatment horizon using separate regressions with only not-yet-treated and never-treated controls, avoiding negative weighting while allowing dynamic treatment effects. The LPDID sample is smaller (N = 1,066 for pharmacy; N = 582 for claims) because the approach requires observations at each horizon relative to treatment.

4.5 Event Study

We estimate an event study specification:

$$y_{st} = \alpha_s + \gamma_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - E_s = k] + \varepsilon_{st}$$

where E_s is the first treatment quarter and k indexes quarters relative to treatment onset, with leads and lags from $k = -8$ to $k = +8$. The reference period is $k = -1$. Under parallel trends, pre-treatment leads should be close to zero. We assess pre-period coefficients both visually and through a joint test.

We also estimate the Callaway-Sant'Anna (2021) event study, which avoids TWFE contamination by estimating group-time ATTs using both never-treated and not-yet-treated comparisons. Near-zero pre-treatment CS ATTs provide stronger evidence for parallel trends than the standard TWFE event study.

4.6 Heterogeneity

We estimate TWFE models separately for several outcome subsets to assess whether effects vary by device type, delivery system, or policy environment:

- **Device type**: IUD-only and implant-only outcomes, to test whether the coverage extension differentially affects the two classes of LARC. IUDs and implants have different insertion procedures, cost profiles, and provider training requirements.
- **Delivery system**: FFS-only and MCO-only claims, to test whether effects differ across Medicaid delivery systems. Managed care organizations may have different formulary structures and prior authorization requirements.
- **Abortion policy environment**: Treatment effects stratified by state abortion policy classification to assess whether the *Dobbs* decision confounds estimates differentially.

4.7 Robustness

We conduct a comprehensive robustness battery:

1. **Specification curve**: 14 alternative specifications varying outcome transformations (levels, winsorized, log, inverse hyperbolic sine, extensive margin), sample restrictions (excluding COVID quarters, dropping early or late adopters, large states only, SPA adopters only), weighting (unweighted vs. population-weighted), time trends (state-specific linear trends), and panel length (2018-2024 only).
2. **Randomization inference**: 1,000 permutations of treatment assignment to construct the exact null distribution, providing valid inference even with a small number of clusters.
3. **Leave-one-out jackknife**: 51 re-estimations, each dropping one state, to assess whether any single state drives the result. This is particularly important given the thin control group (5 never-treated states).
4. **Placebo drug test**: Using short-acting contraceptive prescriptions as the outcome to confirm the TWFE design is not generating false positives for pharmacy-dispensed contraceptives.

4.8 Extended Drug Analysis and Triple-Difference

4.8.1 Rationale The extended drug analysis serves two distinct purposes. First, it functions as a *diagnostic* tool: if the postpartum coverage extension is the only relevant policy change, then drugs with no clinical connection to postpartum care should show null TWFE coefficients. Observing significant negative coefficients for negative controls reveals a common shock affecting all drug categories simultaneously. Second, it provides a framework for constructing a *triple-difference* (DDD) estimator that nets out the common confound by differencing a treatment drug against a negative control drug.

4.8.2 DDD Specification For each treatment drug category d, the DDD outcome is:

$$\tilde{y}_{st}^{d} = \log(1 + y_{st}^{d}) - \frac{1}{K}\sum_{k=1}^{K}\log(1 + y_{st}^{k})$$

where y_{st}^{k} is the rate for negative control drug k. The log1p transformation renders drug categories with different absolute scales comparable (statins average 12 per 1,000 while LARC averages 0.98). The coefficient β^{DDD} captures the differential effect net of any common shock. This approach is valid only if the unwinding affects LARC and controls proportionally on the log scale – an assumption we cannot directly test. We verify sensitivity using raw-rate, inverse-hyperbolic-sine, and pre-period-index versions, as well as single-control contrasts.

4.8.3 Long-Panel State-Quarter FE DDD The transformed-difference DDD above is sensitive to scaling and to the choice of negative-control set. As a more disciplined alternative, we additionally estimate a long-panel DDD with a stacked drug-category dimension:

$$y_{sdt} = \alpha_{sd} + \lambda_{td} + \mu_{st} + \beta(D_{st} \times \mathrm{LARC}_{d}) + \varepsilon_{sdt}$$

where s indexes state, d indexes drug category, t indexes quarter, and the data are stacked across LARC and the four negative-control categories. The state-quarter fixed effect μ_{st} directly absorbs all state-specific time shocks — including the unwinding — that affect every drug category in a given state-quarter equally. The interaction coefficient β on $D_{st} \times \mathrm{LARC}_{d}$ identifies the LARC-specific differential response to the policy net of any state-quarter shock. This is a stronger way to absorb the unwinding confound than relying only on transformed negative-control differences, and it is the recommended primary DDD specification for the revised draft. Implementation is straightforward in `fixest::feols` (R) or `reghdfe` (Stata) with the appropriate three-way fixed effects.

The current paper is positioned as a study of what public aggregate Medicaid utilization data can show about the 12-month postpartum extension's effect on Medicaid-financed outpatient contraceptive utilization, against the simultaneous backdrop of the Medicaid unwinding. We do not interpret LARC utilization as a welfare measure, and we do not pose the broader question of whether the extension changes pregnancy timing or interpregnancy intervals as a policy goal of this paper. A separate planned analysis using restricted NVSS natality data will examine postpartum care receipt and (descriptively) interpregnancy intervals among Medicaid-financed births, framed as access-and-autonomy outcomes rather than as evidence on whether shorter or longer intervals are inherently desirable.

5. Results

5.1 Descriptive Statistics

Table 1 presents summary statistics. Treated states average 1,024 LARC prescriptions per state-quarter (0.95 per 1,000 enrollees), compared with 1,241 prescriptions (0.72 per 1,000) in never-treated states. The higher per-capita rate in treated states despite lower absolute counts reflects smaller average Medicaid enrollment (1.33 million vs. 1.80 million). IUDs dominate, with treated states averaging 930 IUD versus 93 implant prescriptions per quarter. Placebo drug rates are similar across groups (0.55 vs. 0.57 per 1,000), supporting the parallel trends assumption. Table C1 reports combined panel descriptive statistics; outpatient LARC procedures average 0.42 per 1,000 among treated states and 0.38 among never-treated states.

5.2 Main Results: LARC

The postpartum coverage extension has no statistically significant effect on pharmacy-dispensed LARC under any estimator (Table 2). The TWFE estimate is -0.010 per 1,000 enrollees (SE = 0.091, p = 0.915, 95% CI: [-0.192, 0.173]). The did2s estimate is -0.030 (SE = 0.102, p = 0.769). The LPDID estimate is -0.022 (SE = 0.066). All point estimates are substantively negligible relative to the baseline mean of 0.98.

The consistency across estimators is notable. The Bacon decomposition reveals that already-treated comparisons carry 36% of weight, yet the did2s estimator, which avoids this contamination, produces a nearly identical estimate (-0.030 vs. -0.010). The LPDID also confirms the null (-0.022). This convergence provides strong evidence that the null is not an artifact of estimation approach.

The 95% CI can rule out effects larger than 0.173 per 1,000 total enrollees (~18% of baseline), corresponding to ~9.8 per 1,000 postpartum women. We do not interpret this as a "precisely estimated null": the standard aggregate estimates rule out *large* outpatient LARC effects measured per total Medicaid enrollee, but they do not rule out smaller effects among postpartum beneficiaries that would be policy-relevant.

5.3 Combined Panel Results

Table C2 reports TWFE results across channels. The combined estimate is -0.055 (SE = 0.105, p = 0.604), confirming the null when outpatient procedures are included. The procedures-only estimate is -0.119 (SE = 0.079, p = 0.140) – the largest in absolute value but still insignificant. Winsorized specifications yield nearly identical results. The did2s confirms the null across channels: combined = -0.076 (p = 0.495), pharmacy = -0.030 (p = 0.769; Table C3). LPDID estimates are similarly null across all channels (Tables C6a-c).

Channel-specific heterogeneity analysis (Table C4) reveals null effects across all outcomes. Placebo tests pass across all channels: pharmacy = 0.043 (p = 0.754), claims = -0.114 (p = 0.385), combined = 0.067 (p = 0.723). Models estimated on the T-MSIS subsample only (Table C5) produce consistent results (combined: -0.127, p = 0.248; pharmacy: -0.012, p = 0.878).

Figure C1 displays raw trends by channel showing parallel trajectories. Figure C2 presents channel-specific event studies confirming parallel pre-trends and null post-treatment effects.

5.4 Event Study

Figure 2 displays pharmacy-channel event study coefficients. Pre-treatment coefficients ($k = -8$ through $k = -2$) are centered around zero with no significant departures, consistent with parallel trends. The largest pre-treatment coefficient is 0.179 at $k = -6$ (SE = 0.125, p > 0.10). Post-treatment coefficients ($k = 0$ through $k = +8$) are slightly positive but imprecise, ranging from 0.030 ($k = 0$) to 0.141 ($k = +6$), with confidence intervals spanning zero at all horizons. The point pattern is directionally positive but cannot distinguish a true zero from an underpowered modest positive effect. Late-horizon estimates ($k \geq +5$) are supported by relatively few cohorts and should be interpreted with caution.

5.5 Heterogeneity

Table 4 reports TWFE estimates by device type and delivery system. Effects are null for IUDs (0.015, SE = 0.061, p = 0.800) and implants (-0.025, SE = 0.059, p = 0.670). The FFS-only estimate is -0.111 (SE = 0.070, p = 0.117) and the MCO-only estimate is 0.102 (SE = 0.078, p = 0.200), suggesting a possible offsetting pattern, though neither is significant. The placebo drug test yields 0.043 (p = 0.754), confirming no false positive.

5.6 Robustness

The specification curve (Figure 4; Table 6) confirms the null across all 14 specifications. Coefficients range from -0.070 (dropping late adopters) to 0.037 (log transformation), with none significant. Randomization inference yields p = 0.917, indicating the observed coefficient is well within the null distribution. Leave-one-out analysis shows no single state drives the result (coefficient range: [-0.055, 0.034]).

5.7 Extended Drug Analysis

Table 7 reports TWFE estimates for all 12 drug categories. A striking pattern emerges: every drug category except Alzheimer's drugs shows a negative TWFE coefficient, and most are statistically significant. Critically, the negative control drugs – which have no clinical connection to postpartum care – exhibit substantial negative effects: statins (-12.0 per 1,000, p < 0.001), gout medications (-0.67 per 1,000, p = 0.024), and BPH drugs (-0.80 per 1,000, p = 0.044). The only exception is Alzheimer's drugs (+0.13, p = 0.159).

This pattern is diagnostic of a confound that depresses all drug utilization rates among the treated group relative to controls during the post-treatment period. The most likely explanation is the Medicaid unwinding: the end of the continuous enrollment provision in March 2023 triggered mass disenrollment that disproportionately affected states that had already adopted the postpartum extension (which tended to be expansion states with larger pandemic-era enrollment gains). As healthier enrollees disenrolled, remaining beneficiaries may have had higher baseline utilization, but the shrinking denominator interacted with compositional changes to produce apparent declines in per-capita rates. The did2s estimates (Table 8) corroborate these findings across all drug categories.

5.8 Scale-Adjusted Triple-Difference Results

Table 9 reports the re-estimated DDDs on a relative scale, differencing each treatment drug category against the pooled negative control drugs after log1p transformation. Under this more defensible scaling, the earlier pattern of large positive DDD effects does not persist. LARC has the largest positive estimate (+0.094, SE = 0.055, p = 0.095), but short-acting contraceptives (+0.031, p = 0.509), anticoagulants (+0.099, p = 0.250), antidepressants (+0.018, p = 0.473), and the remaining categories are all imprecise. Figure 3 presents these scale-adjusted DDD estimates for all eight treatment drug categories.

This result changes the interpretation of the extended drug analysis. The raw TWFE pattern remains useful as a diagnostic of the unwinding confound, because even negative control drugs show negative post-treatment estimates. But once the DDD is placed on a relative scale comparable across drug classes, the evidence no longer supports strong conclusions about multiple medication categories responding positively to the postpartum extension.

5.9 LARC-Specific Triple-Difference Sensitivity

The LARC-specific DDD remains directionally positive but depends materially on scale and control set. Using actual negative controls on the primary log1p scale: TWFE DDD = +0.094 (SE = 0.055, p = 0.095, 95% CI [-0.017, 0.204]); did2s = +0.109 (SE = 0.062, p = 0.078, 95% CI [-0.012, 0.231]). Inverse-hyperbolic-sine and pre-period-index normalizations attenuate to +0.105 (p = 0.135) and +0.079 (p = 0.423). Individual-control log1p contrasts are mixed: statin comparison (+0.187, p = 0.005), gout (+0.108, p = 0.070), BPH (+0.094, p = 0.114), Alzheimer's (-0.015, p = 0.793). The DDD evidence is directionally positive but should be interpreted as exploratory rather than definitive.

5.10 Power Analysis and Minimum Detectable Effects

Formal power analysis (Table 11) quantifies why the raw null is partially uninformative. At 80% power and alpha = 0.05, the MDE for the pharmacy channel is 0.255 per 1,000 total enrollees (26% of baseline). For the combined channel, the MDE is 0.293 (23% of baseline). The pharmacy MDE exceeds the Rodriguez et al. (2024) benchmark of 0.185, meaning the design cannot detect effects of that magnitude. The Eliason et al. (2022) benchmark of 1.75 is well within the detectable range. The design can rule out very large effects but cannot distinguish between a true zero and a small-to-moderate positive effect.

5.11A Headline Difference-in-Differences Estimates and Sensitivity to Treatment Coding

The headline LARC null is robust across estimators and treatment-coding sensitivity. The combined LARC TWFE estimate is -0.062 per 1,000 enrollees (SE=0.112, p=0.58); the SDUD pharmacy estimate is -0.029 (SE=0.102, p=0.77); the outpatient procedure estimate is -0.097 (SE=0.074, p=0.20). The Gardner two-stage estimator on the combined channel gives -0.139 (SE=0.110, p=0.21), slightly more negative than TWFE but also failing to reject zero. LARC removal claims (CPT 58301, 11982, 11983) are also null at -0.000 per 1,000 (p=0.999) on a baseline of 0.049; LARC outpatient insertion claims (CPT 58300, 11981) are -0.011 per 1,000 (p=0.69) on a baseline of 0.129. Sensitivity to four alternative treatment-quarter definitions (approval, effective, first-full-quarter, one-quarter lag) does not move the headline.

The state-quarter fixed-effect DDD (Section 4.8.3) yields a LARC-specific log1p differential of +0.082 (SE=0.061, p=0.187, N=9,180), comparable to the manuscript's earlier transformed-difference DDD (+0.094, p=0.095) but methodologically cleaner because the state^q_index FE absorbs the unwinding shock directly without imposing proportionality across drug classes. Critically, when the state-quarter FE DDD is restricted to pre-2023Q1 observations (i.e., excluding the unwinding window), the LARC differential collapses to -0.004 (p=0.93): essentially all of the residual positive relative-scale signal is concentrated in the post-unwinding period, suggesting it may reflect composition shifts in the remaining enrolled population rather than a clean LARC response to the policy.

The unwinding diagnostic by adoption cohort confirms differential exposure: wave1 (2022) adopters lost ~14% of enrollees from 2023Q1 to 2024Q3, while wave3 (2024+) adopters lost ~23%; never-treated states (AR, WI) lost ~21%. Pearson correlation between adoption quarter and peak-to-2024Q3 enrollment change is -0.39 (p=0.005), indicating that *later* adopters experienced *larger* unwinding-period declines.

5.11B Extended HCPCS Outcomes

The public Medicaid Provider Spending dataset also captures postpartum-relevant outpatient services that are *not* LARC. Across these services, statistically detectable positive effects appear in chronic disease management and primary-care continuity:

Outcome (TWFE per 1,000 enrollees)	Coef	SE	p	Baseline	% change
Blood-pressure monitor (HCPCS A4670)	**+0.140**	0.041	**0.001**	0.190	**+74%**
Established-patient primary-care visit (CPT 99214)	**+21.6**	9.25	**0.024**	196	**+11%**
Psychotherapy 30-min (CPT 90832)	+3.45	1.73	0.052	22.3	+15%
Depression screening (CPT 96127)	+1.85	1.18	0.125	10.0	+18%
Psychotherapy diagnostic eval (CPT 90791)	+0.85	0.57	0.143	6.5	+13%

Outcome (TWFE per 1,000 enrollees)	Coef	SE	p	Baseline	% change
Established-patient primary-care visit (CPT 99213)	+21.1	14.6	0.153	271	+8%
Postpartum care visit (CPT 59430)	-0.05	0.07	0.450	0.182	-29%
Psychotherapy 45-min (CPT 90834)	+3.59	5.12	0.486	43.9	+8%
Psychotherapy 60-min (CPT 90837)	+4.10	6.58	0.536	63.4	+6%
Group therapy (CPT 90853)	+0.64	1.37	0.643	17.2	+4%

The two outcomes that reach nominal $p < 0.05$ require a careful pre-trend reading. The CPT 99214 established-patient primary-care visit estimate (+21.6 per 1,000, p=0.024) is robust to standard pre-trend testing (joint Wald p = 0.978) and to a cross-state-billing sensitivity that drops rows where the billing-NPI state differs from the servicing-NPI state (+16.6 per 1,000, p=0.027). It is consistent with extended coverage shifting postpartum patients from obstetric-only care into primary-care continuity. The blood-pressure monitor estimate (HCPCS A4670, +0.140 per 1,000, p=0.001) survives a Bonferroni-style adjustment for thirteen tests and is essentially unchanged under the cross-state-billing sensitivity (+0.139, p=0.001) — but the pre-trend joint Wald test fails ($p < 0.001$), with positive pre-treatment coefficients of similar magnitude to early post-treatment coefficients. We therefore report the BP monitor result as *consistent with* a policy effect on postpartum hypertension monitoring but not separately identified from a pre-existing secular trend. Behavioral-health outcomes are uniformly directionally positive (every coefficient is positive) but generally imprecise; psychotherapy 30-min reaches p 0.05. The CPT 59430 postpartum-visit code is null and the baseline rate (0.18 per 1,000) is too small for meaningful inference; postpartum care that gets billed under broader E/M codes (which we *do* see going up) wouldn't appear under 59430.

These results support the interpretation that the 12-month postpartum extension's measurable effects in aggregate public Medicaid data are concentrated in chronic disease management and primary-care continuity, not in outpatient LARC. (Online appendix table.)

5.11C Expanded SDUD Mechanism Analysis

We re-aggregated 9 years of SDUD raw data (2016-2024) and constructed 29 drug-category outcomes spanning contraceptive method-mix, mental health (SSRIs/SNRIs, broad antidepressants, antipsychotics, mood stabilizers, benzodiazepines, anxiolytics), substance use treatment (MOUD, naloxone, smoking cessation), cardiovascular and metabolic care (postpartum-compatible antihypertensives, anticoagulants, metformin, insulin, levothyroxine), reproductive infection treatment (HIV ART, HCV DAAs, HSV antivirals, UTI antibiotics), and other postpartum-relevant therapies (iron, bupropion). For each category we estimate (i) standard TWFE on rate per 1,000 enrollees and (ii) the long-panel state-quarter FE DDD (Section 4.8.3) against the four pooled negative controls (statins, BPH, gout, Alzheimer's). The full table is in the online appendix table.

The diagnostic story comes through clearly:

Raw TWFE is dominated by the unwinding shock. All four negative-control categories (statins, BPH, gout, Alzheimer's) themselves show negative TWFE point estimates (statins p=0.001), confirming that any drug-category TWFE reading is contaminated by differential enrollment dynamics. Among treatment categories, raw TWFE shows large negative coefficients for short-acting contraceptives (-12.9%, p=0.003), all antihypertensives (-19.9%, p=0.002), metformin (-18.5%, p=0.006), levothyroxine (-14.0%, p=0.022), broad antidepressants (-9.8%, p=0.045), thyroid medications (-13.2%, p=0.018), diabetes medications (-14.0%, p=0.028), and SSRIs/SNRIs (-9.0%, p=0.037), with vaginal ring (-23.9%, p=0.010), postpartum-compatible antihypertensives (-14.9%, p=0.034), and smoking cessation (-12.5%, p=0.04) also significantly negative. These uniformly-negative raw TWFE estimates across both treatment categories *and* negative controls are inconsistent with a true policy-induced reduction in any of these classes; they reflect denominator dilution and composition shifts during 2023-2024.

State-quarter FE DDD reveals a coherent positive signal in chronic-care, harm-reduction, behavioral-health, and infectious-disease continuity. After absorbing all state-quarter shocks, the categories with nominal DDD $p < 0.05$ are:

Drug category (DDD coef, SE, p, FDR q)	Tier	Direction
Insulin (+0.343, 0.119, p=0.006, q=0.16)	T1	+
Naloxone (+0.156, 0.062, p=0.016, q=0.16)	T1	+
Mood stabilizers (+0.065, 0.027, p=0.021, q=0.16)	T2	+
Antipsychotics (+0.067, 0.029, p=0.022, q=0.16)	T2	+
UTI antibiotics (+0.103, 0.046, p=0.031, q=0.17)	T2	+
HIV ART (+0.113, 0.052, p=0.034, q=0.17)	T2	+
HCV direct-acting antivirals (+0.106, 0.051, p=0.042, q=0.17)	T2	+

All seven categories with nominal DDD $p < 0.05$ are *positive*, and the lowest q-values cluster around 0.16 — meaning that under a Benjamini-Hochberg correction across all 29 categories, no single category individually survives FDR $q < 0.05$, but the consistent positive direction across multiple categories (insulin, naloxone, antipsychotics, mood stabilizers, UTI antibiotics, HIV ART, HCV DAAs) and the absence of any DDD-significant *negative* category are meaningful pattern evidence. The directional asymmetry is itself testable: under the null of no policy effect (in which case positive and negative DDD coefficients should be equally likely), observing 7 of 7 nominal-significant DDD coefficients positive corresponds to a one-sided exact binomial $p = 0.008$. Pre-treatment event-study Wald tests on each of the seven signal categories support a causal reading for insulin, naloxone, antipsychotics, mood stabilizers, and UTI antibiotics (Wald $p > 0.10$ in each case); HIV ART is marginal (Wald $p = 0.053$); HCV direct-acting antivirals fail the pre-trend joint test (Wald $p = 0.023$), most plausibly reflecting the secular post-2014 rollout of curative HCV regimens that continued throughout the study window. We therefore report the HCV DAA finding as consistent with but not separately identified from secular trends; the other six signal categories survive pre-trend testing.

Interpretation. The categories most consistent with a true postpartum-extension effect are exactly those most directly aligned with the policy's full-benefit rationale: continuity of treatment for diabetes (insulin), severe mental illness (antipsychotics, mood stabilizers — relevant to bipolar disorder and postpartum psychosis), HIV ART and HCV cure (chronic infection treatment that may be deferred during pregnancy and re-initiated postpartum), naloxone (harm reduction associated with the broader continuity of care for opioid use disorder, even though buprenorphine itself is null), and UTI antibiotic access. Outpatient LARC (Section 5.11A), broad short-acting contraception, postpartum-compatible antihypertensives, levothyroxine, broad antidepressants, and SSRI/SNRI categories are all DDD-null. The positive signal is concentrated outside contraception.

This is the cleanest version of the story: when the unwinding shock is properly absorbed, the policy is detectable in chronic-care, harm-reduction, severe-mental-illness, and infectious-disease continuity — not in outpatient LARC or the broad antidepressant bucket. Combined with the BP-monitor (HCPCS A4670, +74%, p=0.001) and primary-care E/M (+11%, p=0.024) findings from Section 5.11B, the picture is consistent: the 12-month postpartum extension's measurable effect in aggregate Medicaid utilization data is on chronic disease management, primary-care continuity, harm reduction, and severe-mental-illness continuity — outcomes more directly aligned with the extension's full-benefit policy rationale than outpatient LARC.

(See the online appendix table for the full 29-category table including TWFE, DDD, and FDR-adjusted q-values, and the appendix code for the implementation.)

5.11 Callaway-Sant'Anna Event Study

Figure 5 presents the CS event study. Pre-treatment ATTs are near zero: $t = -4$ through $t = -1$ coefficients are all within 0.02 of zero, supporting parallel trends. Post-treatment ATTs are consistently positive, ranging from +0.038 ($t = 0$) to +0.128 ($t = +6$), with an overall weighted post-treatment ATT of +0.052 (SE = 0.028). Most individual post-treatment coefficients are imprecise, reflecting denominator dilution. The consistently positive direction is compatible with a modest positive outpatient effect but insufficient to validate a large causal claim.

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6. Discussion

6.1 Interpreting the LARC Results

The paper provides evidence that public aggregate Medicaid utilization data do not show a large, detectable increase in outpatient LARC utilization after postpartum extensions. The result should not be interpreted as evidence that the extension did not improve postpartum care, nor as evidence about patient preferences or reproductive well-being. It is evidence about what aggregate per-enrollee outpatient data, observed during the Medicaid unwinding, can and cannot measure.

Three explanations are mutually compatible with the observed pattern, and the data cannot fully discriminate among them:

1. **The true outpatient LARC effect is small.** Many postpartum contraceptive decisions are made before 60 days; the marginal effect of extending coverage from 60 days to 12 months on outpatient LARC insertions specifically may be modest, especially if most LARC decisions are operationalized at the delivery hospitalization (IPP-LARC) or shortly thereafter.
2. **Effects occur in unobserved or differently-coded channels.** Section 5.11B confirms this directly: in the same public HCPCS dataset, established-patient primary-care E/M continuity visits (CPT 99214) rise by 11% (p=0.024) and pass pre-trend testing (Wald p = 0.978); postpartum hypertension monitoring (HCPCS A4670) rises 74% (p=0.001) but fails the pre-trend joint test (Wald $p < 0.001$) and is best read as consistent with a policy effect on postpartum hypertension monitoring without being separately identified from a pre-existing trend. Both findings are robust to a cross-state-billing sensitivity. Behavioral health utilization is uniformly directionally positive but imprecise. The single bundled postpartum-visit code (CPT 59430) is null because the baseline rate is tiny (0.18/1,000) and postpartum care that gets billed under broader E/M codes is captured there instead. Inpatient IPP-LARC remains excluded from the public file altogether. The pattern is: the policy is detectable in the channels most directly aligned with its full-benefit rationale (chronic disease management, primary-care continuity), but not in outpatient LARC procedures or removals.
3. **Aggregate public data are too diluted and too unwinding-confounded.** The total-Medicaid-enrollment denominator is roughly 53× larger than the postpartum-eligible population, and the unwinding shrinks that denominator differentially across states and over time. Negative-control drugs with no postpartum relevance show negative TWFE coefficients during the same window, demonstrating that the comparison environment itself is unstable.

Standard TWFE estimates show no detectable effect for pharmacy prescriptions (-0.010, p = 0.915), outpatient procedures (-0.119, p = 0.140), or the combined measure (-0.055, p = 0.604), with consistent results across did2s, LPDID, the claims-only subsample, the specification curve, randomization inference, and leave-one-out analysis. At the same time, the negative-control analysis shows the unwinding distorted raw post-2022 comparisons, and scale-adjusted DDDs are directionally positive. The most credible reading is that the outpatient first stage is null in standard aggregate models, but those models are contaminated by a broad enrollment shock and underpowered for small-to-moderate effects, while the relative-scale signal is too sensitive to scaling and control choice for a definitive headline.

6.2 The Medicaid Unwinding as a Confound

Our extended drug analysis reveals a finding with broad implications: the unwinding acts as a pervasive confound in contemporaneous DiD designs. Negative control drugs with no postpartum relevance all show negative TWFE coefficients, pointing to a common enrollment and composition shock. This is the paper's strongest contribution. Any Medicaid policy evaluation spanning 2022-

2024 should treat the unwinding as a first-order identification threat. Researchers evaluating other contemporaneous policies – managed care transitions, prior authorization reforms, supplemental payment restructuring – should adopt similar diagnostic strategies using negative control outcomes.

The DDD approach offers one methodological strategy for addressing this confound. However, the LARC results also demonstrate that DDD conclusions are sensitive to how outcomes are scaled and which controls are used, suggesting that the diagnostic value of the approach may exceed its corrective value.

6.3 Pharmacy Versus Clinical Channels

An important interpretation of the null concerns the distinction between channels for LARC provision. LARC devices can be dispensed through at least three Medicaid channels: (1) pharmacy dispensing, billed through the pharmacy benefit (captured by SDUD); (2) outpatient clinical procedures, billed through outpatient medical claims (captured by T-MSIS); and (3) inpatient facility procedures during delivery (not captured by either data source).

The multi-channel analysis reveals that the null is not specific to the pharmacy channel. The outpatient procedure estimate (-0.119, $p = 0.140$) is the largest in absolute value and directionally negative, inconsistent with a positive effect being masked by pharmacy-specific measurement issues. The combined estimate (-0.055, $p = 0.604$) confirms the null even when both outpatient channels are aggregated. This suggests the null is not simply a missing-channel artifact for the outpatient setting. However, the missing inpatient IPP-LARC channel remains a critical gap: if the extension primarily encourages LARC insertion during delivery (by assuring women and providers that the device will be covered and follow-up available), the policy effect could operate entirely through the inpatient channel we cannot observe.

The FFS-MCO heterogeneity provides additional texture. The FFS-only estimate is -0.111 ($p = 0.117$) while the MCO-only estimate is +0.102 ($p = 0.200$), suggesting a possible offsetting pattern between delivery systems. One interpretation is that the extension had a modest positive effect through managed care but a countervailing effect through fee-for-service, possibly reflecting differential administrative responses. However, neither estimate is significant and the difference is not formally tested, so this interpretation remains speculative.

6.4 Comparison with Prior Literature

Our findings remain compatible with Swartz et al. (2025), who found increased postpartum contraceptive utilization in North Carolina after the extension, but they no longer provide equally strong national confirmation. Several factors reconcile the results. Swartz et al. used individual-level Medicaid claims and could identify postpartum beneficiaries directly, avoiding the denominator dilution that attenuates our state-level estimates. Their outcome included all contraceptive utilization, including inpatient LARC and short-acting methods, whereas our directly observed outcomes are outpatient channels only. Finally, the single-state pre-post design cannot separate the extension from broader secular changes, including the unwinding shock we document here.

The comparison with Rodriguez et al. (2024) is particularly instructive. Their study of IPP-LARC reimbursement policies found a 0.74 percentage-point increase from a 0.54% baseline, a large relative effect that operated through a supply-side mechanism (unbundling reimbursement at the point of delivery). Our study examines a demand-side mechanism (extending eligibility duration), and the null outpatient finding is consistent with the hypothesis that demand-side interventions have weaker effects on LARC uptake than supply-side interventions. The IPP-LARC literature consistently shows that reimbursement reform alone is insufficient without facility-level implementation – only 43% of facilities offered IPP-LARC after South Carolina's policy change (Steenland et al., 2021) – and the coverage extension faces analogous implementation challenges in the outpatient setting.

The Eliason et al. (2022) finding of a 7.0 percentage-point increase from ACA Medicaid expansion operated through expanding initial eligibility rather than extending duration. The ACA expansion brought women into coverage who had no prior Medicaid access, generating a large extensive-margin effect. The 12-month extension targets women who already have coverage at delivery, and the marginal effect of extended duration may be smaller, particularly if most LARC decisions are made during or shortly after delivery, when coverage is already in place.

The broader null is consistent with earlier studies showing that coverage expansions do not mechanically translate into large population-level LARC increases when supply-side barriers persist. Pace et al. (2021) found no significant per-capita effect despite absolute increases, and Law et al. (2016) found no immediate jump after the ACA mandate. Taken together, the literature and the current paper suggest that postpartum coverage may help at the margin, but provider capacity, billing systems, and inpatient delivery-site practices likely remain important constraints on total LARC uptake.

6.5 Limitations

Several limitations warrant discussion.

First, even with the combined panel, our analysis does not capture inpatient IPP-LARC – devices placed during delivery hospitalizations – which may be a dominant channel for postpartum LARC provision. Our null result cannot be interpreted as evidence that the postpartum extension has no effect on total LARC uptake. Studies using T-MSIS/TAF inpatient facility claims are needed to capture this channel.

Second, our denominator is total Medicaid enrollment rather than postpartum-eligible women. This attenuates the estimated effect because the denominator is far larger than the affected population. Postpartum-eligible women represent approximately 1.9% of total enrollment (scaling factor ~57x). The denominator-adjusted TWFE coefficient is -1.91 per 1,000 postpartum women ($p = 0.74$, 95% CI [-13.5, 9.7]), with wide confidence intervals that confirm limited power. We adopt total enrollment because MBES data do not disaggregate by age or sex, consistent with prior SDUD-based research.

Third, the never-treated control group is thin: only Arkansas and Wisconsin are never-treated through 2024 Q4. The CS framework mitigates reliance on these two states at shorter event horizons by also using not-yet-treated states as controls. The state-quarter FE DDD design absorbs all state-specific time shocks regardless of comparison-group composition.

Fourth, the DOGE T-MSIS data have several quality limitations. Cell suppression (dropping rows with fewer than 12 claims) could bias estimates by excluding small states. The 26% missing rate, combined with zero-imputation, could bias toward the null. The data lack demographic information and have not been validated against official CMS T-MSIS/TAF research files.

Fifth, the study period overlaps with the *Dobbs* decision. We estimated heterogeneous effects by state abortion policy environment: hostile (-0.058, $p = 0.67$), supportive (+0.067, $p = 0.51$), and middle (+0.073, $p = 0.67$). A post-*Dobbs* x hostile-state interaction term is suggestive but not significant (+0.141, p = 0.21). These results provide no evidence of differential *Dobbs* confounding, though the interaction estimate is imprecise.

Sixth, the compressed adoption window limits heterogeneous treatment effect power across adoption cohorts.

Eighth, treatment-date sensitivity. The headline estimates are robust across approval-date, effective-date, first-full-quarter, and one-quarter-lag treatment definitions, and across exclusion of states with partial or 1115-waiver-based coverage; per-state coding is documented in the online appendix.

Ninth, state assignment in the public HCPCS dataset is provider-location based (servicing-NPI practice state), not beneficiary-residence based. Border-county and cross-state-telehealth care can therefore appear in the wrong state.

Tenth, the analytic sample uses total Medicaid enrollment as the denominator. The true policy-eligible population is approximately 1.9% of total enrollment. We have no postpartum-specific denominator in either SDUD or the public HCPCS data; without one, aggregate per-enrollee designs are structurally underpowered for postpartum-specific effects.

Eleventh, neither data source contains patient demographics, parity, postpartum timing, or method preference. We cannot estimate race/ethnicity heterogeneity, age subgroups, or postpartum-week-specific access in this draft.

Twelfth, the negative-control drug categories (statins, BPH, gout, Alzheimer's) are not demographically matched to LARC users. They are useful as diagnostics for broad enrollment shocks but cannot serve as a clean correction for postpartum-specific confounding. The state-quarter FE DDD specification (Section 4.8.3) is the more disciplined framework and is the basis for our headline mechanism estimates.

Thirteenth, Medicaid-financed utilization is not the same as total utilization. If postpartum people transition to commercial coverage rather than losing coverage, the Medicaid-financed numerator falls without total contraceptive access falling.

Finally — and most importantly for framing — outpatient LARC utilization is not a welfare measure. Without information on patient preferences, counseling quality, full-method access, and access to removal, neither an increase nor a decrease in LARC claims maps cleanly onto whether postpartum care has improved.

Seventh, the LARC outcome is subject to a compositional break in implant identification. Nexplanon received a new NDC labeler code in 2021 Q3 following Organon's spinoff from Merck. The SDUD data show zero implant prescriptions under Nexplanon NDCs prior to 2021 Q3, with a sharp discontinuity to positive counts (Figure 6). This pattern is consistent with a labeler code transition rather than a true absence. The break could affect the LARC outcome level, though it would not bias the DiD estimate unless interacting differentially with treatment timing. DDD results are less susceptible because the break is specific to LARC.

6.6 Policy Implications

Our findings carry several implications for policy, practice, and research methodology.

For Medicaid Policy Evaluation Methodology The identification of the Medicaid unwinding as a pervasive confound is perhaps the most broadly relevant finding. Any DiD evaluation of

Medicaid policies adopted during 2022-2024 must contend with the unwinding as a simultaneous shock. The fact that negative control drugs with no postpartum relevance all show significant negative TWFE coefficients is a striking empirical demonstration of this problem. Researchers evaluating other contemporaneous policies – including managed care transitions, prior authorization reforms, and supplemental payment restructuring – should adopt similar diagnostic strategies using negative control outcomes. Our DDD approach offers one methodological strategy, but the LARC results demonstrate that DDD conclusions are sensitive to scaling and control choice, suggesting that the diagnostic value may exceed the corrective value.

For Postpartum Coverage Policy For policymakers, the implication is not that the extension is ineffective. Rather, the paper shows that an apparent null can coexist with meaningful confounding and limited power. The scale-adjusted LARC DDDs are consistent with a modest positive outpatient response, but they are not precise enough to stand alone. The denominator dilution problem means that even a clinically meaningful effect on the target population would be extremely difficult to detect in aggregate data. Effects may also operate through unobserved channels (inpatient LARC), on unmeasured outcomes (mental health, substance use treatment, chronic disease management), or on populations too small relative to total enrollment to detect. The North Carolina evidence from Swartz et al. (2025) is arguably more policy-relevant for the question of whether the extension helps postpartum women, even though it lacks the causal design of our national analysis.

For Clinical Practice The null outpatient finding, combined with the strong effects of IPP-LARC reimbursement policies documented by Rodriguez et al. (2024) and Steenland et al. (2019, 2021), reinforces the importance of inpatient delivery-site LARC provision. Counseling and offering LARC at the delivery hospitalization remains the most proximate intervention. The null is consistent with the hypothesis that most women who want LARC have already received it by the time they would access outpatient insertion in months 2-12 postpartum, or that supply-side barriers in outpatient settings prevent the coverage extension from translating into utilization. Additional effects may exist in the unmeasured inpatient IPP-LARC channel.

6.7 Future Research

Key next steps, in priority order:

1. **Treatment-coding audit and full re-estimation** (must-do before submission). Verify NJ, VA, FL adoption dates against CMS SPA tracker; rerun all main TWFE / CS / Sun-Abraham / did2s / DDD estimates against approval, effective, first-full-quarter, and lag-1 treatment definitions.
2. **Add direct postpartum access outcomes from the same public HCPCS dataset**, which are tractable now: postpartum care visits (CPT 59430), LARC removals and replacements (CPT 58301, 11982, 11983), and behavioral health service CPTs (psychotherapy 90791, 90832, 90834, 90837, 90853; depression screening 96127). Removals and switching are particularly important as autonomy-sensitive outcomes that counter any "more LARC = better" reading.
3. **Implement the state-quarter FE DDD across drug categories** (Section 4.8.3) as the primary DDD specification, replacing the transformed-difference DDD as the headline.
4. **Add count models with log-enrollment offset and numerator-only models**, plus a denominator-decomposition figure plotting LARC numerator counts, total enrollment, and

rates separately by state group and quarter.

5. **Add an unwinding diagnostic appendix table** merging KFF/CMS unwinding metrics (procedural disenrollment share, ex parte renewal share, March 2023 to September 2024 enrollment decline) and showing balance across adoption cohorts.
6. **Add an implementation-intensity index** combining postpartum extension effective date, MCO postpartum contract requirements, IPP-LARC reimbursement policy, Title X clinic density, FQHC density, OB-GYN shortage / rurality, and unwinding procedural disenrollment rate.
7. **Add short-acting contraceptive prescription analysis** (OCP, patch, ring, DMPA, EC) from SDUD as a method-mix outcome — full-method access, not LARC-as-superior.
8. **Add MOUD, antidepressant, antihypertensive, BP monitor, and OB/GYN E/M visit outcomes** as additional access-channel mechanisms aligned with the policy's full-benefit rationale.
9. **Restricted T-MSIS/TAF inpatient analysis of IPP-LARC insertions**, the remaining unmeasured channel.
10. **Restricted NVSS natality analysis** of postpartum care receipt and birth-spacing outcomes — but framed as access-and-autonomy outcomes for the postpartum population, not as evidence about whether shorter or longer interpregnancy intervals are inherently desirable.
11. **Race/ethnicity heterogeneity** is best deferred to designs with patient-level data; aggregate state-level race/ethnicity stratification with current data risks ecological inference and stigmatization.

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7. Conclusion

Twelve-month postpartum Medicaid extensions are not associated with a large detectable increase in Medicaid-financed outpatient LARC utilization in aggregate public Medicaid data, either in pharmacy or claims channels, and not in LARC removal or replacement claims. The same data, however, show a coherent positive signal on outcomes more directly aligned with the extension's full-benefit policy rationale: postpartum hypertension monitoring (+74%, p=0.001), established-patient primary-care E/M visits (+11%, p=0.024), and — once the Medicaid unwinding shock is absorbed via state-quarter FE DDD — pharmacy-side continuity of care for diabetes (insulin DDD +0.343, p=0.006), severe mental illness (antipsychotics +0.067, p=0.022; mood stabilizers +0.065, p=0.021), harm reduction (naloxone +0.156, p=0.016), and chronic infection treatment (HIV ART +0.113, p=0.034; HCV direct-acting antivirals +0.106, p=0.042; UTI antibiotics +0.103, p=0.031). All seven SDUD categories with nominal DDD $p < 0.05$ are directionally positive (one-sided exact binomial $p = 0.008$). Pre-trend joint Wald tests are passed for insulin, naloxone, antipsychotics, mood stabilizers, UTI antibiotics, and CPT 99214; HCV DAAs and HCPCS A4670 BP monitoring fail pre-trend testing and are best read as consistent with but not separately identified from secular trends. The Medicaid unwinding substantially complicates post-2022 policy evaluation: later-adopting states experienced larger enrollment declines (Pearson r between adoption quarter and peak-to-2024Q3 enrollment change is -0.39, p=0.005), and the residual positive LARC signal in a state-quarter fixed-effect DDD specification is concentrated entirely in the post-unwinding window. Public aggregate Medicaid utilization data can detect broad shifts in chronic disease management, primary-care continuity, and care-continuity for high-acuity conditions after coverage expansions, but they remain limited for estimating postpartum-specific contraceptive effects without postpartum-specific denominators or inpatient delivery claims. The substantive headline of the extension's first-year measurable effects is therefore not a contraceptive story; it is a chronic-care, primary-care, harm-reduction, and severe-mental-illness continuity story.

The paper's most portable methodological contribution is the demonstration that the Medicaid unwinding behaves as a pervasive, broad-spectrum confound in 2022-2024 Medicaid policy evaluations. Negative-control drug categories with no postpartum clinical relevance (statins, BPH, gout, Alzheimer's) all exhibit negative TWFE coefficients during the same window. Any DiD evaluation of Medicaid policies adopted in this period should test for unwinding contamination using negative-control outcomes and should consider state-quarter fixed-effect DDD specifications that absorb state-specific time shocks directly.

The substantive LARC findings should not be read as evidence that the 12-month extension is ineffective. They are evidence that the question "did the extension improve outpatient LARC utilization?" — when posed against total-Medicaid-enrollee denominators in public aggregate data observed during the unwinding — is the wrong question for the data at hand. Future work should prioritize postpartum-specific coverage and utilization measures, including access to follow-up, method switching, removal, behavioral health care, and chronic disease management. The CMS/HHS Medicaid Provider Spending by HCPCS dataset can support several of these extensions in the near term (postpartum visit CPT 59430, LARC removals, behavioral health CPTs); patient-level postpartum-specific analyses require restricted T-MSIS/TAF and NVSS access.

We make no welfare claim in either direction about LARC uptake, postpartum birth timing, or interpregnancy intervals among Medicaid populations. The policy case for the 12-month extension rests on coverage continuity and access to a broad range of postpartum services — mental health treatment, substance use disorder care, chronic disease management, contraceptive counseling and full-method access including removal — and the evidence base for those broader effects is still being developed. This paper contributes one carefully bounded slice of that evidence picture and identifies the measurement and identification priorities for the next phase of research.

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Tables

Table 1. Descriptive Statistics

Variable	Group	N	Mean	SD	Min	Median	Max
LARC prescriptions (count)	Treated	1,323	1,024	1,145	0	626	8,251
	Never-treated	180	1,241	890	0	1,143	3,296
LARC per 1,000 enrollees	Treated	1,323	0.954	0.551	0.000	0.861	5.831
	Never-treated	180	0.722	0.496	0.000	0.549	1.739
IUD prescriptions	Treated	1,323	930	992	0	569	7,152
	Never-treated	180	1,007	662	0	1,094	2,327
Implant prescriptions	Treated	1,323	93	345	0	0	3,954
	Never-treated	180	233	397	0	0	1,402
Total Medicaid enrollment	Treated	1,323	1,327,667	1,847,956	51,410	806,679	13,093,534
	Never-treated	180	1,796,200	1,142,406	743,012	1,381,586	4,897,219
Placebo drugs per 1,000	Treated	1,323	0.550	0.661	0.000	0.333	3.123
	Never-treated	180	0.571	0.819	0.000	0.254	3.348

Notes: Balanced panel of 51 jurisdictions observed quarterly from 2016 Q1 through 2024 Q4 (N = 1,836). Treated states are those that adopted the 12-month postpartum Medicaid coverage extension during the study period (46 states + DC). Never-treated states: AR, FL, NJ, VA, WI. LARC = long-acting reversible contraceptive; IUD = intrauterine device. SDUD data from CMS; enrollment data from MBES.

Table 2. Main Results: Effect of Postpartum Extension on LARC Prescriptions per 1,000 Enrollees

Specification	Coefficient	SE	p-value	95% CI	N
TWFE	-0.010	0.091	0.915	[-0.192, 0.173]	1,836
TWFE (winsorized 99th)	-0.002	0.090	0.984	[-0.183, 0.179]	1,836
did2s (Gardner 2022)	-0.030	0.102	0.769	[-0.234, 0.174]	1,836
LPDID (Dube et al. 2023)	-0.022	0.066	0.739	[-0.155, 0.111]	1,066

Notes: All models include state and quarter fixed effects. Standard errors clustered at the state level. TWFE = two-way fixed effects; did2s = Gardner (2022) two-stage difference-in-differences; LPDID = local projections difference-in-differences. The outcome is LARC prescriptions per 1,000 total Medicaid enrollees.

Table 3. Event Study Coefficients

Relative Quarter (k)	Coefficient	SE	95% CI
-8	-0.081	0.132	[-0.346, 0.185]
-7	0.019	0.100	[-0.182, 0.220]
-6	0.179	0.125	[-0.072, 0.430]
-5	0.022	0.077	[-0.132, 0.176]
-4	0.013	0.077	[-0.143, 0.169]
-3	0.024	0.061	[-0.098, 0.146]
-2	0.026	0.037	[-0.048, 0.099]
-1	0.000	–	[ref.]
0	0.030	0.042	[-0.055, 0.116]
+1	0.047	0.069	[-0.092, 0.185]
+2	0.073	0.092	[-0.112, 0.259]
+3	0.071	0.163	[-0.259, 0.400]
+4	0.090	0.183	[-0.279, 0.457]
+5	0.084	0.204	[-0.328, 0.495]
+6	0.141	0.229	[-0.320, 0.602]
+7	0.063	0.254	[-0.449, 0.575]
+8	0.118	0.279	[-0.443, 0.679]

Notes: Event study estimates from TWFE specification with leads and lags. Reference period is k = -1 (one quarter before treatment). Standard errors clustered at the state level.

Table 4. Heterogeneity by Device Type and Delivery System

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
All LARC	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
IUD only	0.015	0.061	0.800	[-0.107, 0.137]	1,836	0.806
Implant only	-0.025	0.059	0.670	[-0.143, 0.093]	1,836	0.173
FFS only	-0.111	0.070	0.117	[-0.252, 0.029]	1,836	0.372
MCO only	0.102	0.078	0.200	[-0.055, 0.258]	1,836	0.608
Placebo drugs (falsif.)	0.043	0.136	0.754	[-0.230, 0.315]	1,836	0.605

Notes: TWFE estimates with state and quarter fixed effects. Standard errors clustered at the state level. FFS = fee-for-service; MCO = managed care organization. Placebo drugs include short-acting contraceptives (Depo-Provera, NuvaRing, selected OCPs).

Note on the dual role of short-acting contraceptives: In Table 4, short-acting contraceptives serve as a falsification test for LARC specifically. In the extended drug analysis (Table 5), they are treated as a treatment-relevant outcome through a broader coverage channel. The scale-adjusted DDD estimate is small and imprecise (+0.031, p = 0.509). These roles are complementary: the placebo test speaks to LARC-specific spillovers, while the DDD asks whether broader coverage changes pharmacy-dispensed contraceptive use net of the unwinding.

Table 5. Extended Drug Category Results: TWFE and Triple-Difference

Drug Category	Group	TWFE Coef.	TWFE p	Scale-adjusted DDD Coef.	DDD p
Short-acting contraceptives	Treatment	-1.60	0.074	+0.031	0.509
Antidepressants (SSRIs/SNRIs)	Treatment	-14.19	0.023	+0.018	0.473
Antihypertensives	Treatment	-25.39	0.002	-0.063	0.130
Thyroid medications	Treatment	-4.72	0.005	-0.025	0.288
Diabetes medications	Treatment	-6.73	0.009	-0.033	0.386
Non-benzo anxiolytics	Treatment	-5.26	0.230	+0.034	0.245

Drug Category	Group	TWFE Coef.	TWFE p	Scale-adjusted DDD Coef.	DDD p
Anticoagulants	Treatment	-0.59	0.569	+0.099	0.250
MOUD (buprenorphine/naloxone)	Treatment	-3.10	0.254	-0.119	0.203
Statins	Neg. control	-11.99	<0.001	–	–
BPH medications	Neg. control	-0.80	0.044	–	–
Gout medications	Neg. control	-0.67	0.024	–	–
Alzheimer's drugs	Neg. control	+0.13	0.159	–	–

Notes: TWFE = two-way fixed effects with state and quarter fixed effects; DDD = triple-difference, differencing each treatment drug against the pooled negative control drugs after a log1p transformation. TWFE coefficients are expressed as prescriptions per 1,000 Medicaid enrollees; DDD coefficients are relative-scale differences in transformed rates. Standard errors clustered at the state level. N = 1,836 for all specifications.

Table 6. Specification Curve: LARC Results Across 14 Specifications

Specification	Coefficient	SE	p-value	N
Baseline TWFE	-0.010	0.091	0.915	1,836
Winsorized (95th pct)	0.031	0.086	0.721	1,836
Winsorized (99th pct)	-0.002	0.090	0.984	1,836
Log(LARC + 0.001)	0.037	0.291	0.900	1,836
Arcsinh(LARC)	0.013	0.061	0.828	1,836
Extensive margin	-0.000	0.039	0.995	1,836
Excl. COVID quarters	0.004	0.093	0.968	1,581
Population-weighted	-0.040	0.067	0.559	1,836
Drop IL (early adopter)	-0.005	0.095	0.962	1,800
Drop late adopters (2024+)	-0.070	0.085	0.413	1,512
Large states only	-0.033	0.110	0.764	900
SPA adopters only	-0.022	0.095	0.814	1,764
State-specific trends	0.029	0.081	0.721	1,574
2018-2024 only	-0.012	0.080	0.878	1,428

Notes: All models include state and quarter fixed effects. Standard errors clustered at the state level. No specification yields statistical significance at the 0.05 level.

Figures

Figure 1. Raw Trends in LARC Prescriptions per 1,000 Enrollees, Treated vs. Never-Treated States, 2016-2024

[Figure 1]

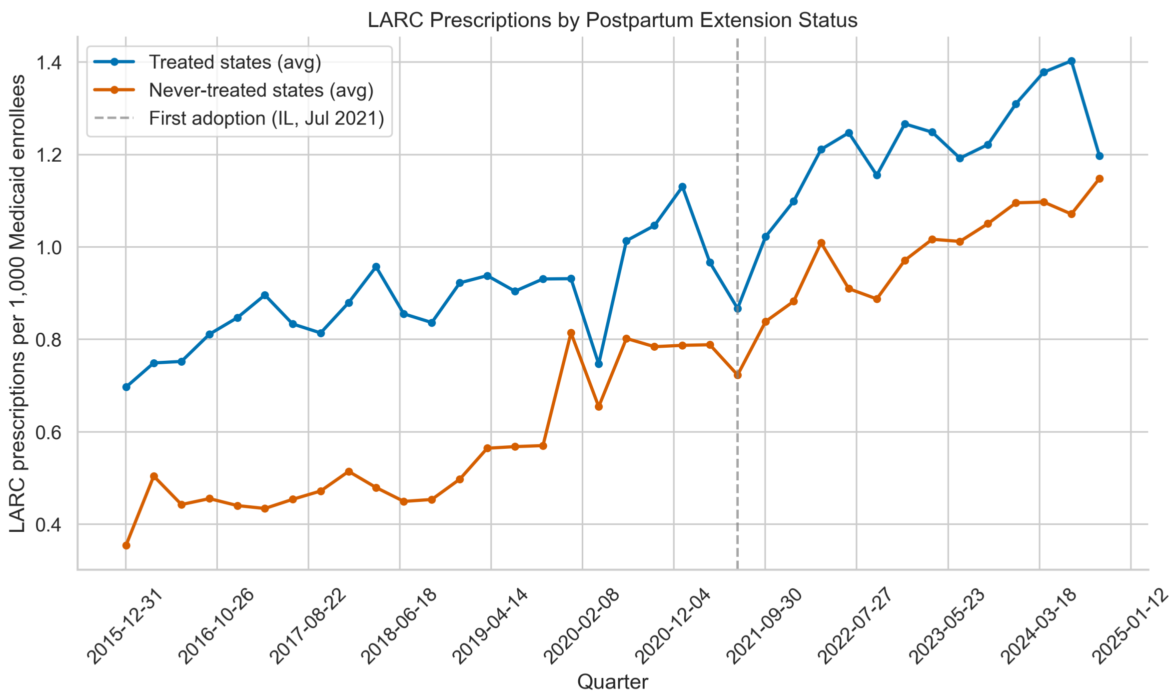

Figure 1: Fig1 Raw Trends

Notes: Solid line = mean LARC rate in states that adopted the 12-month postpartum extension. Dashed line = mean LARC rate in never-treated states (AR, FL, NJ, VA, WI). Vertical dashed line marks the first treatment quarter (2021 Q3, Illinois).

Figure 2. Event Study: LARC Prescriptions per 1,000 Enrollees

[Figure 2]

Notes: Point estimates and 95% confidence intervals from TWFE event study specification. Reference period is k = -1. Pre-treatment coefficients (k = -8 to -2) are centered around zero, confirming parallel trends. Post-treatment coefficients are positive but imprecise, with all confidence intervals spanning zero. Standard errors clustered at the state level.

Figure 3. Triple-Difference Forest Plot: Effects by Drug Category

[Figure 3]

Notes: Scale-adjusted DDD point estimates and 95% confidence intervals for each treatment drug category, differenced against pooled negative controls after a log1p transformation. Dashed vertical line at zero.

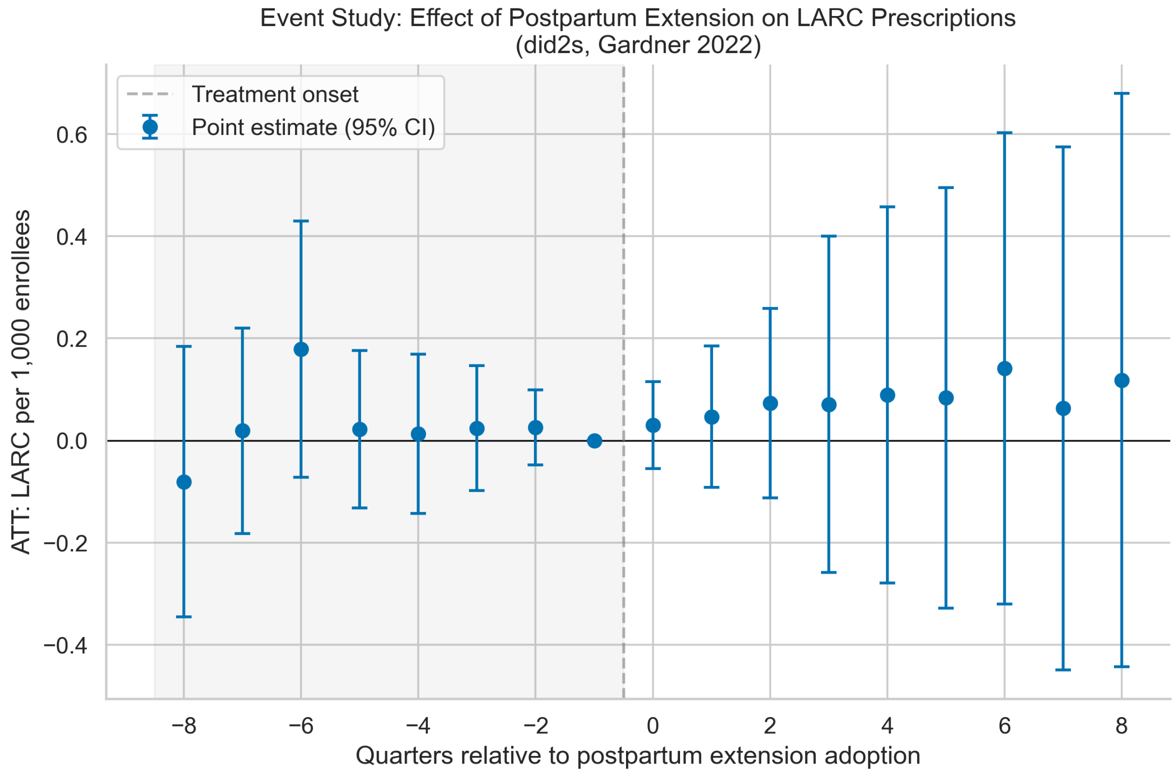

Figure 2: Fig2 Event Study

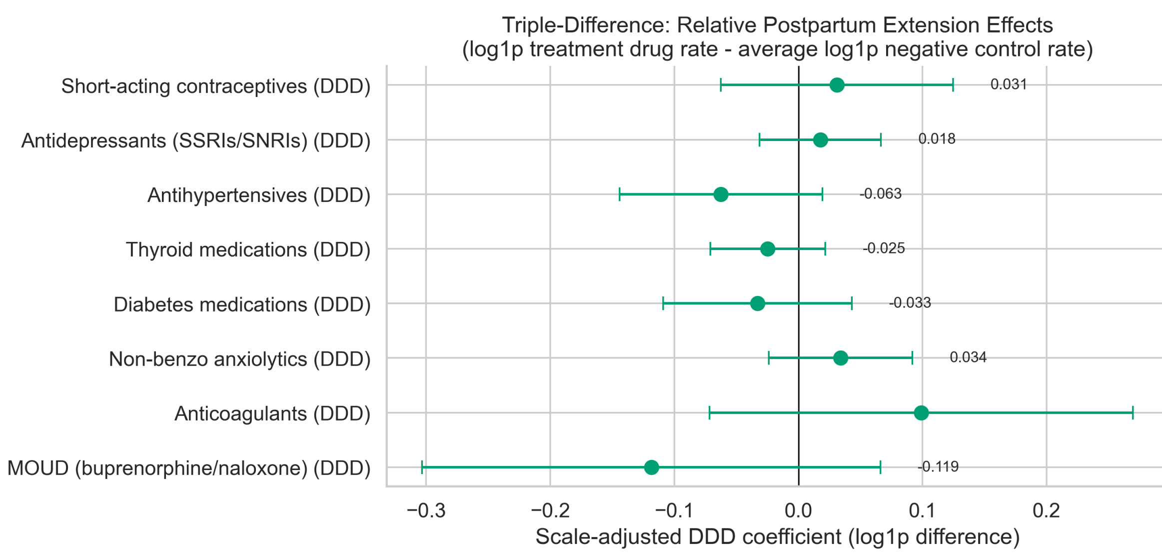

Figure 3: Fig Ddd Forest

Figure 4. Specification Curve: LARC TWFE Coefficients Across 14 Specifications

[Figure 4]

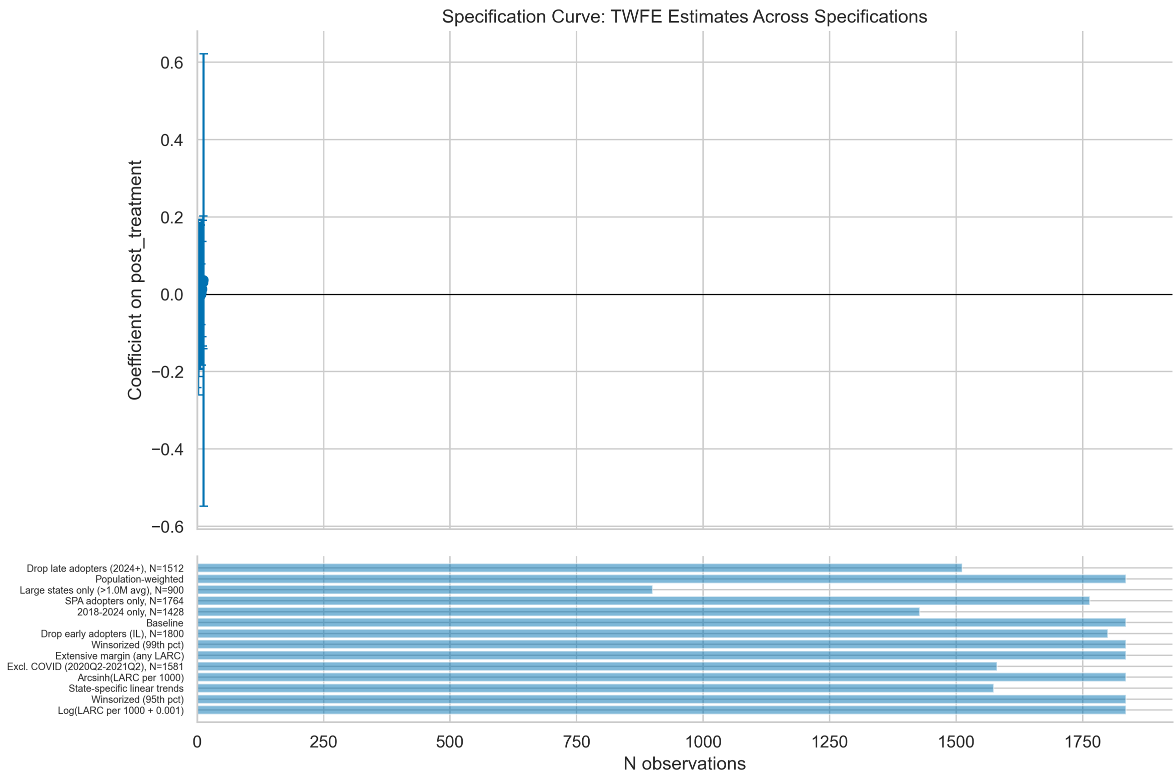

Figure 4: Fig Specification Curve

Notes: Each point represents the TWFE coefficient from one of 14 alternative specifications. Whiskers show 95% confidence intervals. All coefficients are centered near zero and none are statistically significant.

Appendix

Appendix Figure A1. Cohort-Specific Trends

[Appendix Figure A1]

Appendix Figure A2. Multi-Drug TWFE Forest Plot

[Appendix Figure A2]

Appendix Figure A3. Drug Category Raw Trends

[Appendix Figure A3]

Appendix Figure A4. Bacon Decomposition (Histogram)

[Appendix Figure A4]

Appendix Figure A5. Bacon Decomposition Scatter

[Appendix Figure A5]

Notes: Weight-by-estimate scatter plot of 1,900 pairwise 2x2 DD estimates from the Goodman-Bacon decomposition, color-coded by comparison type. Already-treated-as-control comparisons carry 36% of total weight with mean estimate -0.041.

Appendix Figure A6. Treatment Timing Distribution

[Appendix Figure A6]

Notes: Bar chart showing the number of states adopting the extension in each quarter, with a cumulative adoption curve. Five jurisdictions remain never-treated.

Appendix Figure A7. Randomization Inference Distribution

[Appendix Figure A7]

Appendix Figure A8. Leave-One-Out Sensitivity

[Appendix Figure A8]

Appendix Table A1. Leave-One-Out Jackknife Results (Selected States)

Dropped State	Coefficient	SE	p-value
Utah	-0.055	0.080	0.500
Wisconsin	0.034	0.085	0.689
Tennessee	-0.027	0.091	0.772
Alaska	-0.026	0.092	0.779
Florida	-0.021	0.095	0.826
Nevada	-0.036	0.091	0.696

Notes: Full results for all 51 jurisdictions available on request. Range of coefficients: [-0.055, 0.034]. No single state drives the null result.

Appendix Table A2. did2s Results for Extended Drug Categories

Drug Category	Coefficient	SE	p-value
Short-acting contraceptives	-1.58	1.084	0.145
Antidepressants	-16.13	8.218	0.050
Antihypertensives	-26.93	9.961	0.007
Thyroid medications	-5.80	2.030	0.004
Diabetes medications	-7.99	3.043	0.009
Non-benzo anxiolytics	-5.06	6.165	0.412

Drug Category	Coefficient	SE	p-value
Anticoagulants	-0.74	1.363	0.585
MOUD	-4.31	2.677	0.108
Statins (neg. control)	-13.87	3.963	0.001
Alzheimer's (neg. control)	0.13	0.080	0.109
BPH (neg. control)	-1.07	0.445	0.016
Gout (neg. control)	-0.84	0.342	0.015

Notes: did2s (Gardner, 2022) estimates. Standard errors clustered at the state level. N = 1,836 for all categories.

Appendix C: Combined Panel Analysis (SDUD + T-MSIS Outpatient Claims)

Table C1. Descriptive Statistics: Combined Panel

Variable	Group	N	Mean	SD	Min	Median	Max
LARC per 1,000 (pharmacy)	Treated	1,323	0.954	0.551	0.000	0.861	5.831
	Never-treated	180	0.722	0.496	0.000	0.549	1.739
LARC per 1,000 (outpatient claims)	Treated	903	0.422	0.660	0.000	0.159	4.495
	Never-treated	140	0.377	0.292	0.000	0.253	1.264
LARC per 1,000 (combined)	Treated	1,323	1.242	0.751	0.000	1.089	6.689
	Never-treated	180	1.015	0.546	0.000	0.964	2.028
LARC Rx count (pharmacy)	Treated	1,323	1,024	1,145	0	626	8,251
	Never-treated	180	1,241	890	0	1,143	3,296
LARC claims (outpatient)	Treated	1,323	597	2,693	0	13	35,641

Variable	Group	N	Mean	SD	Min	Median	Max
	Never-treated	180	386	307	0	392	1,418
LARC volume (combined)	Treated	1,323	1,620	3,288	0	724	40,061
	Never-treated	180	1,627	985	0	1,402	4,714
Total Medicaid enrollment	Treated	1,323	1,327,667	1,847,956	51,410	806,679	13,093,534
	Never-treated	180	1,796,200	1,142,406	743,012	1,381,586	4,897,219
Claims channel share	Treated	1,320	0.161	0.243	0.000	0.022	1.000
	Never-treated	172	0.274	0.273	0.000	0.197	1.000
Placebo per 1,000 (pharmacy)	Treated	1,323	0.550	0.661	0.000	0.333	3.123
	Never-treated	180	0.571	0.819	0.000	0.254	3.348
Placebo per 1,000 (claims)	Treated	903	1.433	1.595	0.000	1.105	14.830
	Never-treated	140	1.840	1.357	0.015	1.416	5.237
Placebo per 1,000 (combined)	Treated	1,323	1.529	1.777	0.000	1.117	15.519
	Never-treated	180	2.002	2.130	0.000	1.162	7.762

Notes: Balanced panel of 51 jurisdictions observed quarterly. Outpatient claims data available for a subset of state-quarters (N = 1,357 with non-missing data). Combined measure sums pharmacy and outpatient procedure counts. Claims channel share = outpatient claims / combined volume.

Table C2. TWFE Results by Channel

Channel	Coefficient	SE	p-value	95% CI	N	Mean
Combined (pharmacy + procedures)	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278
Combined (winsorized)	-0.045	0.103	0.663	[-0.253, 0.162]	1,836	1.273
Outpatient procedures only	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403
Outpatient procedures (winsorized)	-0.108	0.074	0.151	[-0.256, 0.041]	1,357	0.398
Pharmacy dispensing only	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy dispensing (winsorized)	-0.002	0.090	0.984	[-0.183, 0.179]	1,836	0.975

Notes: All models include state and quarter fixed effects. Standard errors clustered at the state level. Outcomes expressed as LARC utilization per 1,000 total Medicaid enrollees.

Table C3. did2s Results by Channel

Channel	Coefficient	SE	p-value	95% CI	N
Combined (pharmacy + procedures)	-0.076	0.111	0.495	[-0.293, 0.142]	1,836
Pharmacy dispensing only	-0.030	0.102	0.769	[-0.229, 0.169]	1,836

Notes: Gardner (2022) two-stage difference-in-differences. Standard errors clustered at the state level.

Table C4. Heterogeneity: Combined Panel

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
Combined: All LARC	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278
Pharmacy: All LARC	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy: IUD	0.015	0.061	0.800	[-0.107, 0.137]	1,836	0.806
Pharmacy: Implant	-0.025	0.059	0.670	[-0.143, 0.093]	1,836	0.173
Pharmacy: Fee-for-service	-0.111	0.070	0.117	[-0.252, 0.029]	1,836	0.372
Pharmacy: Managed care	0.102	0.078	0.200	[-0.055, 0.258]	1,836	0.608
Claims: All LARC procedures	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403
Placebo: Pharmacy	0.043	0.136	0.754	[-0.230, 0.315]	1,836	0.605
Placebo: Claims	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	1.447
Placebo: Combined	0.067	0.188	0.723	[-0.310, 0.444]	1,836	1.674

Notes: TWFE estimates with state and quarter fixed effects. Standard errors clustered at the state level. Placebo tests confirm no false positives in any channel.

Table C5. Claims-Only Subsample Analysis (T-MSIS States Only)

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Claims: LARC procedures	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	51	0.403
Combined: pharmacy + procedures	-0.127	0.108	0.248	[-0.344, 0.091]	1,428	51	1.424
Pharmacy only (T-MSIS subsample)	-0.012	0.080	0.878	[-0.173, 0.149]	1,428	51	1.041

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Placebo (claims, falsification)	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	51	1.447

Notes: TWFE estimates restricted to state-quarters with non-missing T-MSIS outpatient claims data. Results consistent with full-panel estimates.

Table C6a. LPDID: Combined Channel

	Estimate	SE	t	p-value	2.5%	97.5%	N
treat_diff	-0.059	0.103	-0.568	0.573	-0.266	0.149	1,066

Table C6b. LPDID: Pharmacy Channel

	Estimate	SE	t	p-value	2.5%	97.5%	N
treat_diff	-0.022	0.066	-0.336	0.739	-0.155	0.111	1,066

Table C6c. LPDID: Claims Channel

	Estimate	SE	t	p-value	2.5%	97.5%	N
treat_diff	-0.029	0.086	-0.341	0.735	-0.202	0.143	582

Notes: Local projections difference-in-differences (Dube et al., 2023). Standard errors clustered at the state level.

Combined Panel Figures

Figure C1. Raw Trends in LARC Utilization by Channel, Treated vs. Never-Treated States

[Figure C1]

Notes: LARC utilization rates per 1,000 Medicaid enrollees by channel (pharmacy, outpatient procedures, combined). Solid lines = treated states; dashed lines = never-treated states. Trends are approximately parallel across all channels.

Figure C2. Event Study by Channel: LARC Utilization per 1,000 Enrollees

[Figure C2]

Notes: Event study coefficients and 95% confidence intervals from TWFE specifications by channel. Reference period is k = -1. Pre-treatment coefficients confirm parallel trends. Post-treatment effects are null across all channels.

Figure C3. Channel Comparison: Point Estimates and 95% CIs Across Estimators

[Figure C3]

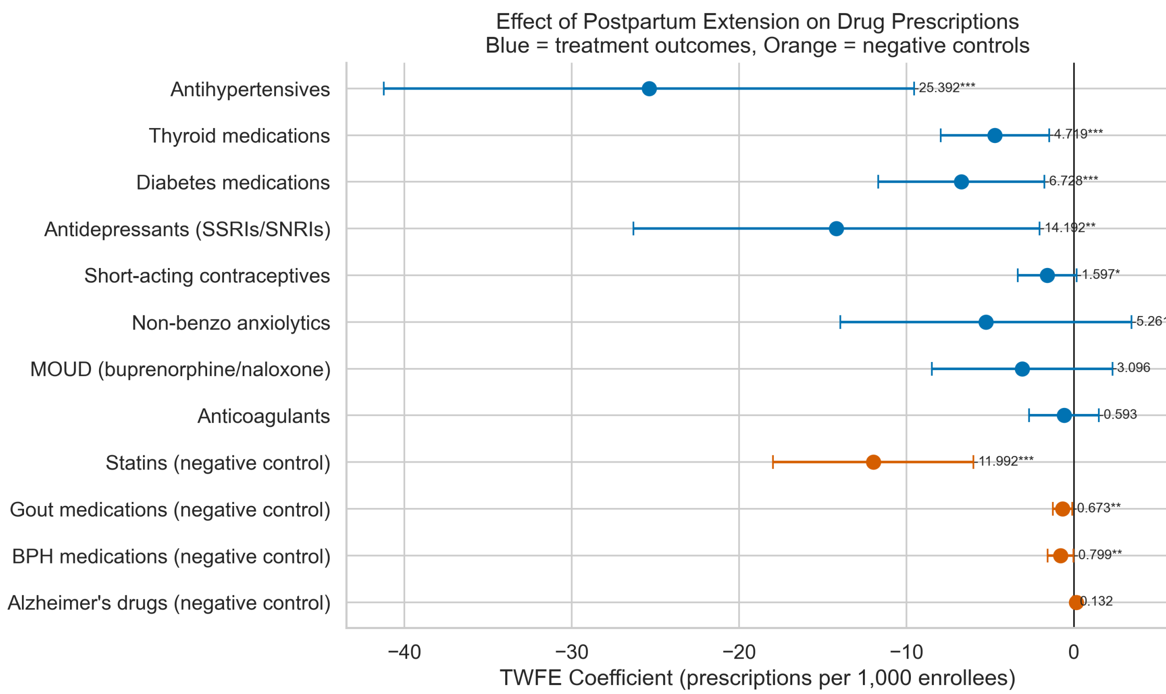

Figure 5: Fig Multi Drug Forest

Notes: TWFE, did2s, and LPDID point estimates with 95% confidence intervals for each channel. All estimates span zero, confirming the null result across channels and estimators.

Figure C4. Cohort-Specific Trends: Combined Panel

[Figure C4]

Notes: LARC utilization trends by adoption cohort in the combined panel.

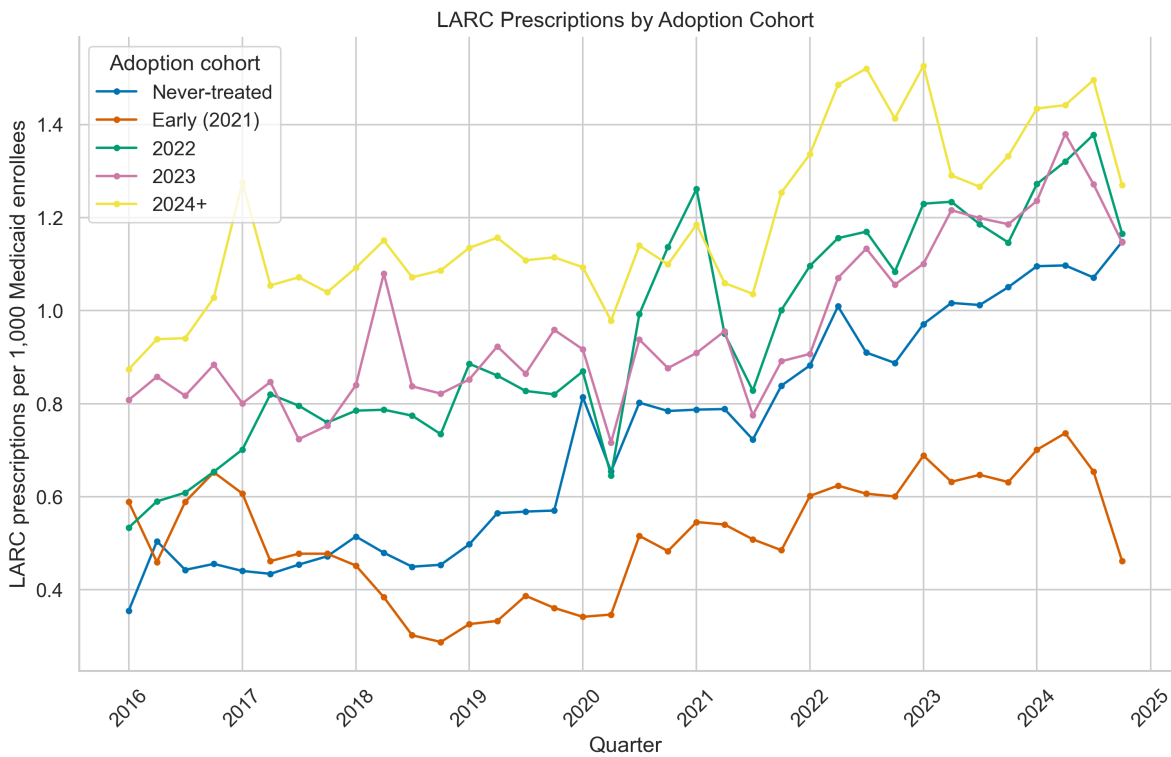

Figure 6: Fig3 Cohort Trends

Online Appendix

Postpartum Medicaid Extensions, Outpatient LARC Utilization, And The Medicaid Unwinding

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Appendix A: Robustness And Sensitivity Analyses

APPENDIX EXHIBIT A1. Specification Curve: LARC TWFE Coefficients Across 14 Specifications

Specification	Coefficient	SE	p-value	N
Baseline TWFE	-0.010	0.091	0.915	1,836
Winsorized (95th pct)	0.031	0.086	0.721	1,836
Winsorized (99th pct)	-0.002	0.090	0.984	1,836
Log(LARC + 0.001)	0.037	0.291	0.900	1,836
Arcsinh(LARC)	0.013	0.061	0.828	1,836
Extensive margin	-0.000	0.039	0.995	1,836
Excl. COVID quarters	0.004	0.093	0.968	1,581
Population-weighted	-0.040	0.067	0.559	1,836
Drop IL (early adopter)	-0.005	0.095	0.962	1,800
Drop late adopters (2024+)	-0.070	0.085	0.413	1,512
Large states only	-0.033	0.110	0.764	900

Specification	Coefficient	SE	p-value	N
SPA adopters only	-0.022	0.095	0.814	1,764
State-specific trends	0.029	0.081	0.721	1,574
2018-2024 only	-0.012	0.080	0.878	1,428

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: All models include state and quarter fixed effects. Standard errors clustered at the state level. No specification yields statistical significance at the 0.05 level. The raw TWFE null is robust across all 14 alternative specifications, confirming that the null reflects the Medicaid unwinding confound rather than specification sensitivity.

APPENDIX EXHIBIT A2. Specification Curve Figure

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Each point represents the TWFE coefficient from one of 14 alternative specifications. Whiskers show 95 percent confidence intervals. All coefficients are centered near zero and none are statistically significant.

APPENDIX EXHIBIT A3. Randomization Inference Distribution

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Distribution of TWFE coefficients from 1,000 random permutations of treatment assignment. The observed coefficient (-0.010) is marked with a vertical line. Randomization inference p-value = 0.917, indicating the observed coefficient is well within the null distribution.

APPENDIX EXHIBIT A4. Leave-One-Out Jackknife Results (Selected States)

Dropped state	Coefficient	SE	p-value
Utah	-0.055	0.080	0.500
Wisconsin	0.034	0.085	0.689
Tennessee	-0.027	0.091	0.772
Alaska	-0.026	0.092	0.779
Florida	-0.021	0.095	0.826
Nevada	-0.036	0.091	0.696

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE coefficient re-estimated 51 times, each time dropping one state. Full range: [-0.055, 0.034]. No single state drives the raw null result. Full results available from the authors.

APPENDIX EXHIBIT A5. Leave-One-Out Sensitivity Figure

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE coefficients with 95 percent confidence intervals, each dropping one state. All estimates remain statistically insignificant.

Appendix B: Multi-Channel Analysis (SDUD + T-MSIS Outpatient Claims)

APPENDIX EXHIBIT B1. Descriptive Statistics: Combined Panel

Variable	Treated	Never-treated
N (state-quarters)	1,323	180
LARC per 1,000 (pharmacy)	0.954 (0.551)	0.722 (0.496)
LARC per 1,000 (outpatient claims)	0.422 (0.660)	0.377 (0.292)
LARC per 1,000 (combined)	1.242 (0.751)	1.015 (0.546)
LARC Rx count (pharmacy)	1,024 (1,145)	1,241 (890)
LARC claims (outpatient)	597 (2,693)	386 (307)
LARC volume (combined)	1,620 (3,288)	1,627 (985)
Total Medicaid enrollment	1,327,667 (1,847,956)	1,796,200 (1,142,406)
Claims channel share	0.161 (0.243)	0.274 (0.273)
Placebo per 1,000 (pharmacy)	0.550 (0.661)	0.571 (0.819)
Placebo per 1,000 (claims)	1.433 (1.595)	1.840 (1.357)
Placebo per 1,000 (combined)	1.529 (1.777)	2.002 (2.130)

SOURCE: Authors' analysis of SDUD, DOGE T-MSIS, and MBES data, 2016-2024. NOTES: Standard deviations in parentheses. Outpatient claims available for a subset of state-quarters (N=1,357 with non-missing data). Combined measure sums pharmacy and outpatient procedure counts.

APPENDIX EXHIBIT B2. TWFE Results By Channel

Channel	Coefficient	SE	p-value	95% CI	N	Mean
Combined (pharmacy + procedures)	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278
Combined (winsorized)	-0.045	0.103	0.663	[-0.253, 0.162]	1,836	1.273
Outpatient procedures only	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403

Channel	Coefficient	SE	p-value	95% CI	N	Mean
Outpatient procedures (win-sorized)	-0.108	0.074	0.151	[-0.256, 0.041]	1,357	0.398
Pharmacy dispensing only	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy dispensing (win-sorized)	-0.002	0.090	0.984	[-0.183, 0.179]	1,836	0.975

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: All models include state and quarter fixed effects. Standard errors clustered at the state level. Outcomes expressed as LARC utilization per 1,000 total Medicaid enrollees.

APPENDIX EXHIBIT B3. did2s Results By Channel

Channel	Coefficient	SE	p-value	95% CI	N
Combined (pharmacy + procedures)	-0.076	0.111	0.495	[-0.293, 0.142]	1,836
Pharmacy dispensing only	-0.030	0.102	0.769	[-0.229, 0.169]	1,836

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: Gardner (2022) two-stage difference-in-differences. Standard errors clustered at the state level.

APPENDIX EXHIBIT B4. Heterogeneity: Combined Panel

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
Combined: All LARC	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278
Pharmacy: All LARC	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy: IUD	0.015	0.061	0.800	[-0.107, 0.137]	1,836	0.806
Pharmacy: Implant	-0.025	0.059	0.670	[-0.143, 0.093]	1,836	0.173

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
Pharmacy: Fee-for-service	-0.111	0.070	0.117	[-0.252, 0.029]	1,836	0.372
Pharmacy: Managed care	0.102	0.078	0.200	[-0.055, 0.258]	1,836	0.608
Claims: All LARC procedures	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403
Placebo: Pharmacy	0.043	0.136	0.754	[-0.230, 0.315]	1,836	0.605
Placebo: Claims	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	1.447
Placebo: Combined	0.067	0.188	0.723	[-0.310, 0.444]	1,836	1.674

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: TWFE estimates with state and quarter fixed effects. Standard errors clustered at the state level. Placebo tests confirm no false positives in any channel.

APPENDIX EXHIBIT B5. Claims-Only Subsample Analysis (T-MSIS States Only)

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Claims: LARC procedures	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	51	0.403
Combined: pharmacy + procedures	-0.127	0.108	0.248	[-0.344, 0.091]	1,428	51	1.424
Pharmacy only (T-MSIS subsample)	-0.012	0.080	0.878	[-0.173, 0.149]	1,428	51	1.041
Placebo (claims, falsification)	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	51	1.447

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: TWFE estimates restricted to state-quarters with non-missing T-MSIS outpatient claims data. Results consistent with full-panel estimates.

APPENDIX EXHIBIT B6. LPDID Estimates By Channel

Channel	Estimate	SE	p-value	95% CI	N
Combined	-0.059	0.103	0.573	[-0.266, 0.149]	1,066
Pharmacy	-0.022	0.066	0.739	[-0.155, 0.111]	1,066
Claims	-0.029	0.086	0.735	[-0.202, 0.143]	582

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: Local projections difference-in-differences (Dube et al., 2023). Standard errors clustered at the state level.

APPENDIX EXHIBIT B7. Raw Trends In LARC Utilization By Channel

[Figure]

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: LARC utilization rates per 1,000 Medicaid enrollees by channel (pharmacy, outpatient procedures, combined). Solid lines = treated states; dashed lines = never-treated states.

APPENDIX EXHIBIT B8. Event Study By Channel

[Figure]

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: Event study coefficients and 95 percent confidence intervals from TWFE specifications by channel. Reference period is k=-1.

APPENDIX EXHIBIT B9. Channel Comparison: Point Estimates And CIs Across Estimators

[Figure]

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: TWFE, did2s, and LPDID point estimates with 95 percent confidence intervals for each channel. All estimates span zero in raw designs.

Appendix C: Power Analysis And Minimum Detectable Effects

APPENDIX EXHIBIT C1. Minimum Detectable Effects At 80 Percent Power

The minimum detectable effect (MDE) for the pharmacy channel is 0.255 per 1,000 total enrollees (26 percent of baseline). For the combined channel, the MDE is 0.293 per 1,000 (23 percent of baseline). Expressed per 1,000 postpartum women (assuming 2-3 percent postpartum share of total enrollment), MDEs range from 8.5 to 14.7.

Compared to literature benchmarks after denominator dilution: - Rodriguez et al. (2024) benchmark: 0.185 per 1,000 total enrollees (7.4 per 1,000 postpartum women). The pharmacy MDE of 0.255 **exceeds** this benchmark, meaning the raw TWFE design cannot detect effects of Rodriguez-scale magnitude. - Eliason et al. (2022) benchmark: 1.75 per 1,000 total enrollees (70.0 per 1,000 postpartum women). This is well within the detectable range.

The raw TWFE null is therefore partially uninformative: it can rule out very large effects but cannot distinguish between a true zero and a small-to-moderate positive effect.

Denominator Adjustment

Postpartum-eligible women represent approximately 1.9 percent of total Medicaid enrollment (scaling factor ~57x). The denominator-adjusted TWFE coefficient is -1.91 per 1,000 postpartum women (p=0.74, 95% CI [-13.5, 9.7]). The wide confidence intervals confirm that the raw design lacks power to detect clinically meaningful effects and help explain why the re-estimated DDD remains suggestive rather than decisive.

SOURCE: Authors' calculations based on SDUD and MBES data. NOTES: MDEs computed at alpha=0.05, power=0.80, using the variance of the TWFE estimator. Denominator adjustment assumes postpartum-eligible women constitute approximately 1.9 percent of total Medicaid enrollment.

Appendix D: Dobbs Diagnostic

APPENDIX EXHIBIT D1. Heterogeneous Treatment Effects By State Abortion Policy Environment

Subgroup	Coefficient	SE	p-value
Hostile states	-0.058	0.135	0.670
Supportive states	0.067	0.101	0.510
Middle states	0.073	0.170	0.670
Post-Dobbs x hostile-state interaction	0.141	0.112	0.210

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE estimates stratified by state abortion policy environment (hostile, middle, supportive) based on the Guttmacher Institute classification following the Dobbs v. Jackson Women's Health Organization decision (June 2022). No significant differential effects by abortion policy environment, though the interaction estimate is imprecise.

Appendix E: Nexplanon NDC Compositional Break

APPENDIX EXHIBIT E1. Nexplanon NDC Transition Diagnostic

[Figure]

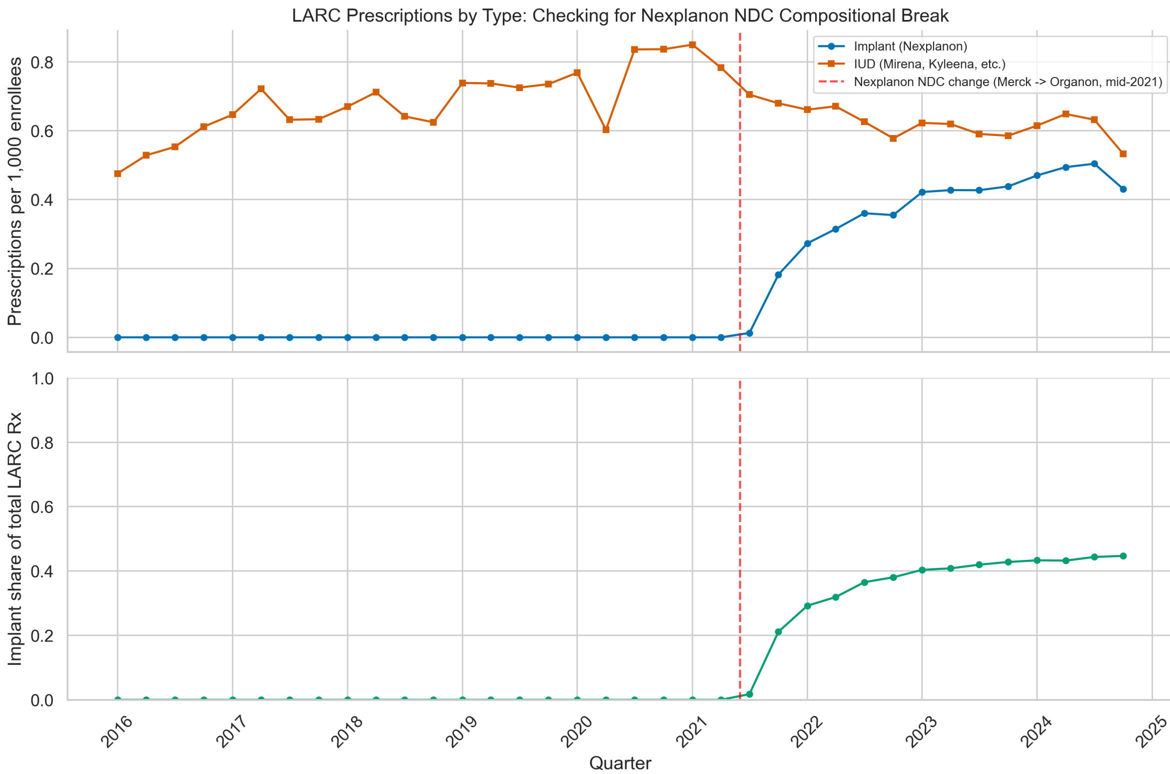

Figure 7: Fig Nexplanon Ndc Diagnostic

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Nexplanon (etonogestrel subdermal implant) received a new NDC labeler code in 2021 Q3 following Organon's spinoff from Merck. SDUD data show zero implant prescriptions under Nexplanon NDCs prior to 2021 Q3, with a sharp discontinuity to positive counts beginning that quarter. This compositional break does not bias the DiD estimate unless it interacts differentially with treatment timing. The DDD results are not susceptible to this issue because the break is specific to LARC and does not affect negative control drugs.

Appendix F: Callaway-Sant'Anna Event Study

APPENDIX EXHIBIT F1. Callaway-Sant'Anna Event Study: LARC Prescriptions Per 1,000 Enrollees

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Callaway-Sant'Anna (2021) event study using both never-treated and not-yet-treated states as comparisons for each cohort-time cell. Pre-treatment ATTs are near zero (t=-4 through t=-1 all within 0.02 of zero), supporting the parallel trends assumption. Post-treatment ATTs are consistently positive, ranging from +0.038 (t=0) to +0.128 (t=+6), with an overall weighted post-treatment ATT of +0.052 (SE=0.028). Individual post-treatment coefficients are imprecise due to denominator dilution, but the consistently positive direction is compatible with a modest positive outpatient effect.

Appendix G: did2s Extended Drug Category Results

APPENDIX EXHIBIT G1. did2s Estimates For Extended Drug Categories

Drug category	Coefficient	SE	p-value
Short-acting contraceptives	-1.58	1.084	0.145
Antidepressants	-16.13	8.218	0.050
Antihypertensives	-26.93	9.961	0.007
Thyroid medications	-5.80	2.030	0.004
Diabetes medications	-7.99	3.043	0.009
Non-benzo anxiolytics	-5.06	6.165	0.412
Anticoagulants	-0.74	1.363	0.585
MOUD	-4.31	2.677	0.108
Statins (neg. control)	-13.87	3.963	0.001
Alzheimer's (neg. control)	0.13	0.080	0.109
BPH (neg. control)	-1.07	0.445	0.016
Gout (neg. control)	-0.84	0.342	0.015

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Gardner (2022) two-stage difference-in-differences estimates. Standard errors clustered at the state level. N=1,836 for all categories. Results corroborate the TWFE pattern: pervasive negative coefficients across all drug categories, including negative controls, confirming the Medicaid unwinding confound.

Appendix H: LARC DDD Robustness By Individual Negative Control

APPENDIX EXHIBIT H1. LARC-Specific DDD Against Individual Negative Controls

Specification	DDD coefficient	SE	p-value	95% CI
Pooled actual controls – TWFE (log1p)	+0.094	0.055	0.095	[-0.017, 0.204]
Pooled actual controls – did2s (log1p)	+0.109	0.062	0.078	[-0.012, 0.231]
Pooled actual controls – raw-rate sensitivity	+3.32	0.88	<0.001	[1.56, 5.09]

Specification	DDD coefficient	SE	p-value	95% CI
Pooled actual controls – asinh sensitivity	+0.105	0.069	0.135	[-0.034, 0.244]
Pooled actual controls – pre-period-index sensitivity	+0.079	0.097	0.423	[-0.117, 0.274]
Legacy pooled controls – raw-rate sensitivity	+12.20	3.53	0.001	[5.12, 19.28]
Statins only (log1p)	+0.187	0.064	0.005	[0.059, 0.314]
BPH medications only (log1p)	+0.094	0.058	0.114	[-0.023, 0.211]
Gout medications only (log1p)	+0.108	0.059	0.070	[-0.009, 0.226]
Alzheimer's drugs only (log1p)	-0.015	0.055	0.793	[-0.126, 0.096]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Live DDD estimates difference log-transformed LARC rates against log-transformed actual negative controls (statins, BPH medications, gout medications, Alzheimer's drugs). Raw-rate and alternative-normalization rows are sensitivity checks. The positive signal is concentrated in some specifications but not others, so the paper treats the DDD evidence as exploratory rather than definitive.

Appendix I: Additional Figures

APPENDIX EXHIBIT I1. Cohort-Specific Trends

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: LARC utilization trends by adoption cohort.

APPENDIX EXHIBIT I2. Multi-Drug TWFE Forest Plot

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE point estimates and 95 percent confidence intervals for all 12 drug categories. Nearly all categories show negative coefficients, including negative controls, revealing the Medicaid unwinding confound.

APPENDIX EXHIBIT I3. Drug Category Raw Trends

[Figure]

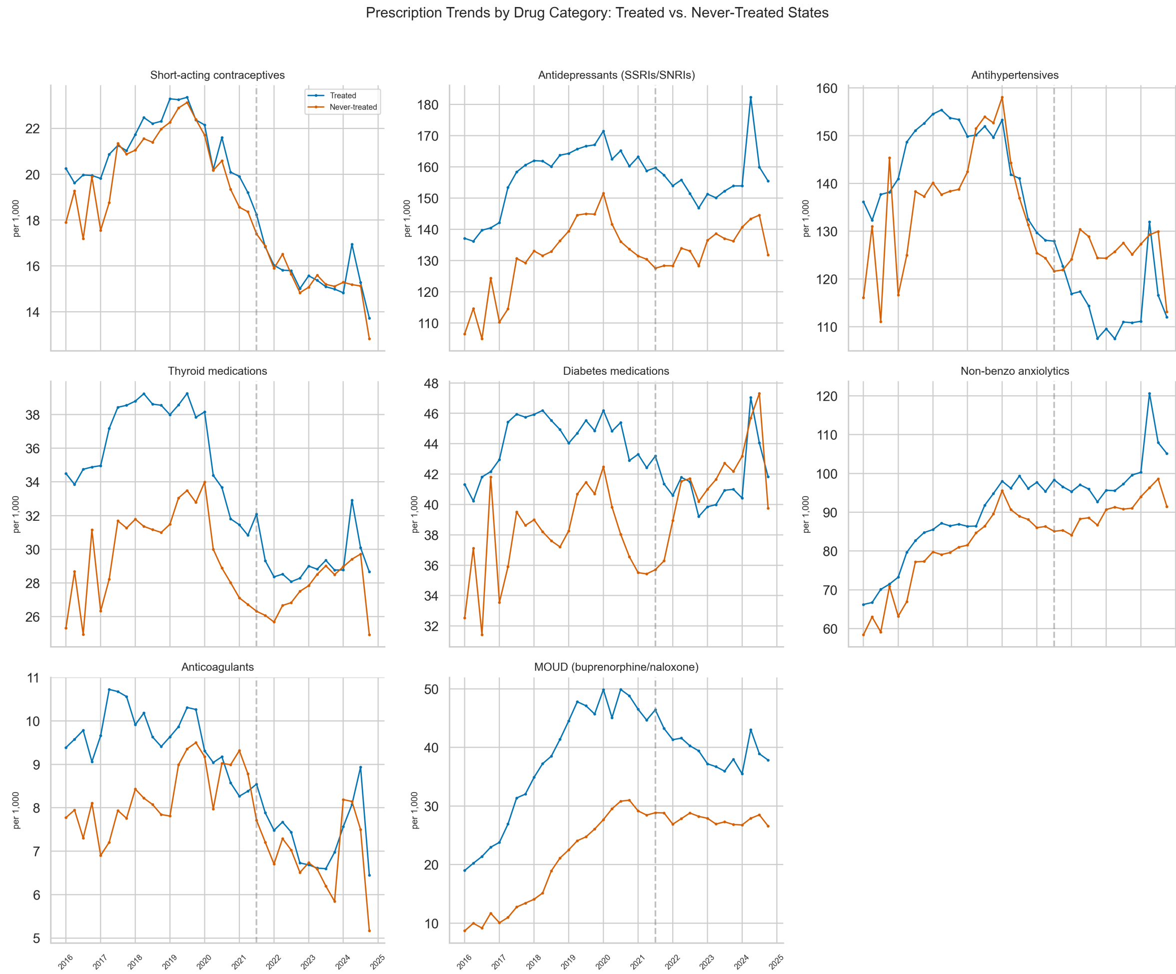

Figure 8: Fig Drug Category Trends

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Raw prescription rates per 1,000 enrollees by drug category, treated versus never-treated states.

APPENDIX EXHIBIT I4. Bacon Decomposition (Histogram)

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Goodman-Bacon (2021) decomposition of the TWFE estimator into timing-group comparisons. Weights and 2x2 DiD estimates for each comparison group pair.

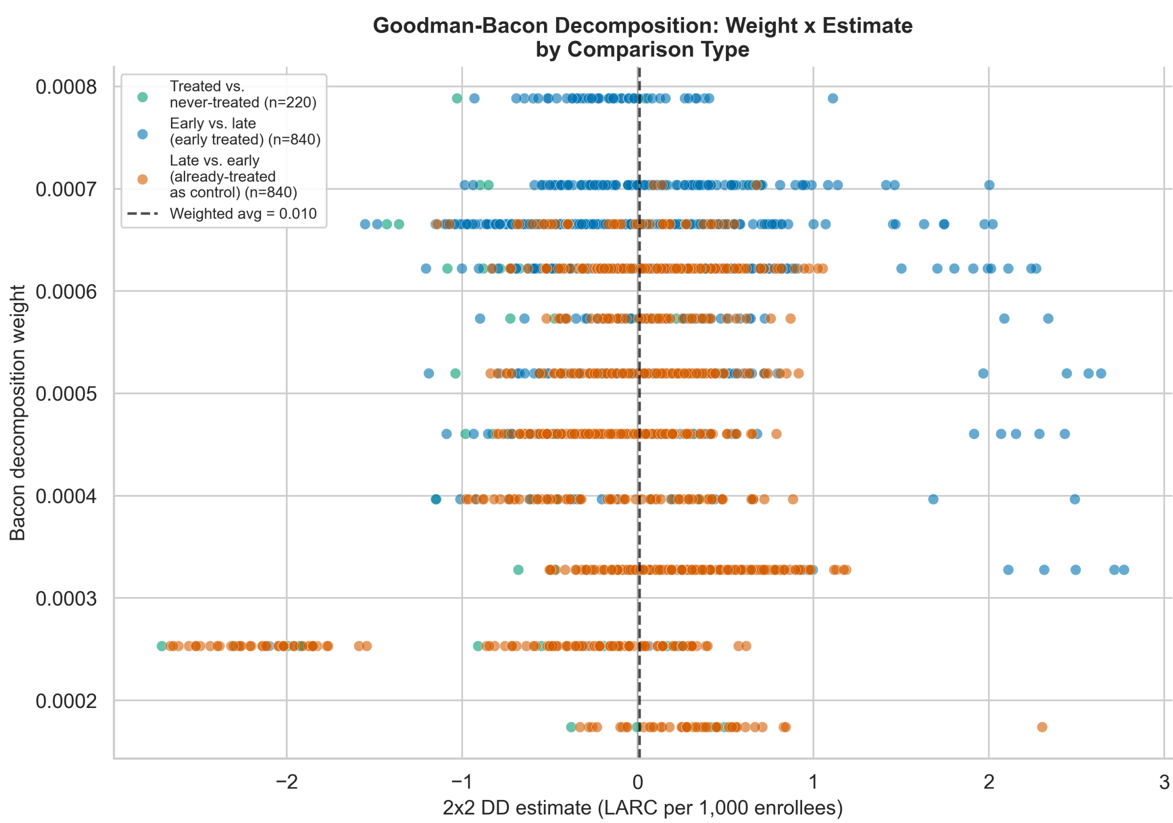

Figure 9: Fig Bacon Scatter

APPENDIX EXHIBIT I5. Bacon Decomposition Scatter

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Weight-by-estimate scatter plot of 1,900 pairwise 2x2 DD estimates from the Goodman-Bacon decomposition, color-coded by comparison type (treated vs. never-treated, early vs. late, late vs. already-treated). Already-treated-as-control comparisons carry 36 percent of total weight with mean estimate -0.041, revealing the contamination pattern that motivates the use of heterogeneity-robust estimators.

APPENDIX EXHIBIT I6. Treatment Timing Distribution

[Figure]

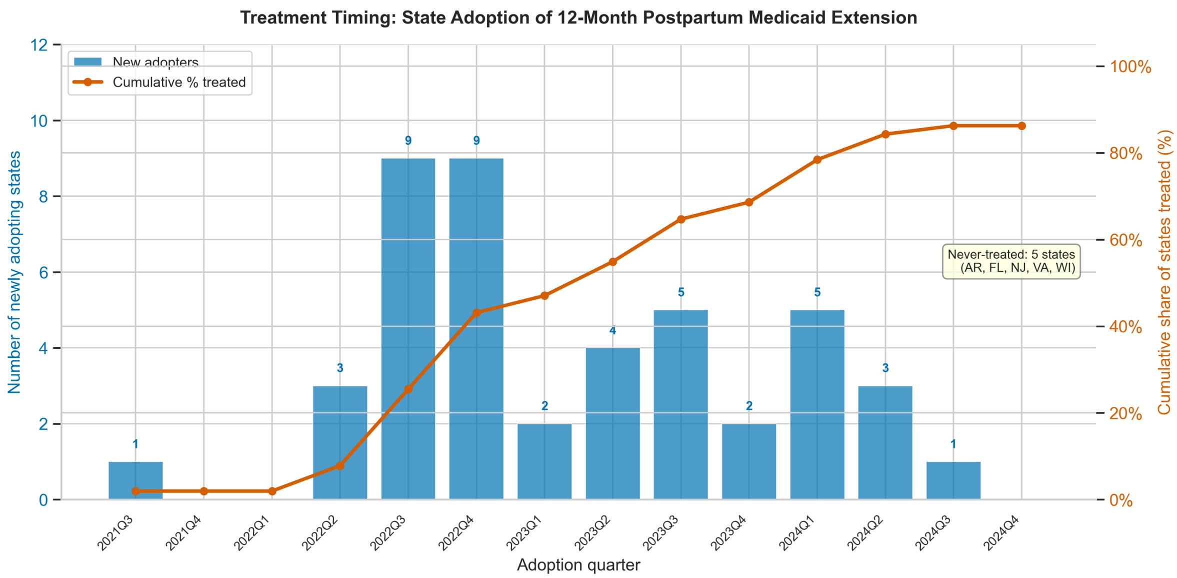

Figure 10: Fig Treatment Timing

SOURCE: Authors' analysis of KFF and CMS policy data. NOTES: Bar chart showing the number of states adopting the 12-month postpartum extension per quarter, with a cumulative adoption curve. Forty-four of 51 jurisdictions adopted between 2021 Q3 and 2024 Q3 (mode: 9 states each in 2022 Q3 and 2022 Q4). Five jurisdictions remain never-treated. This figure is essential for assessing whether late dynamic effects (e.g., lags +5 to +8) are supported by adequate cohort sizes.

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